

Commvault Professional

Course Guide

Module 2 – Storage Configuration

Commvault Platform Release 2023E

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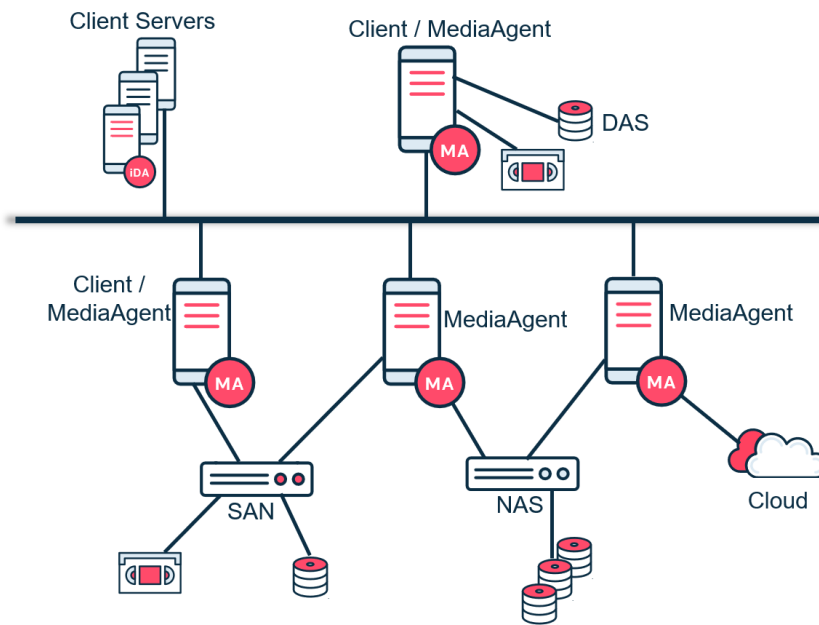
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Storage Overview



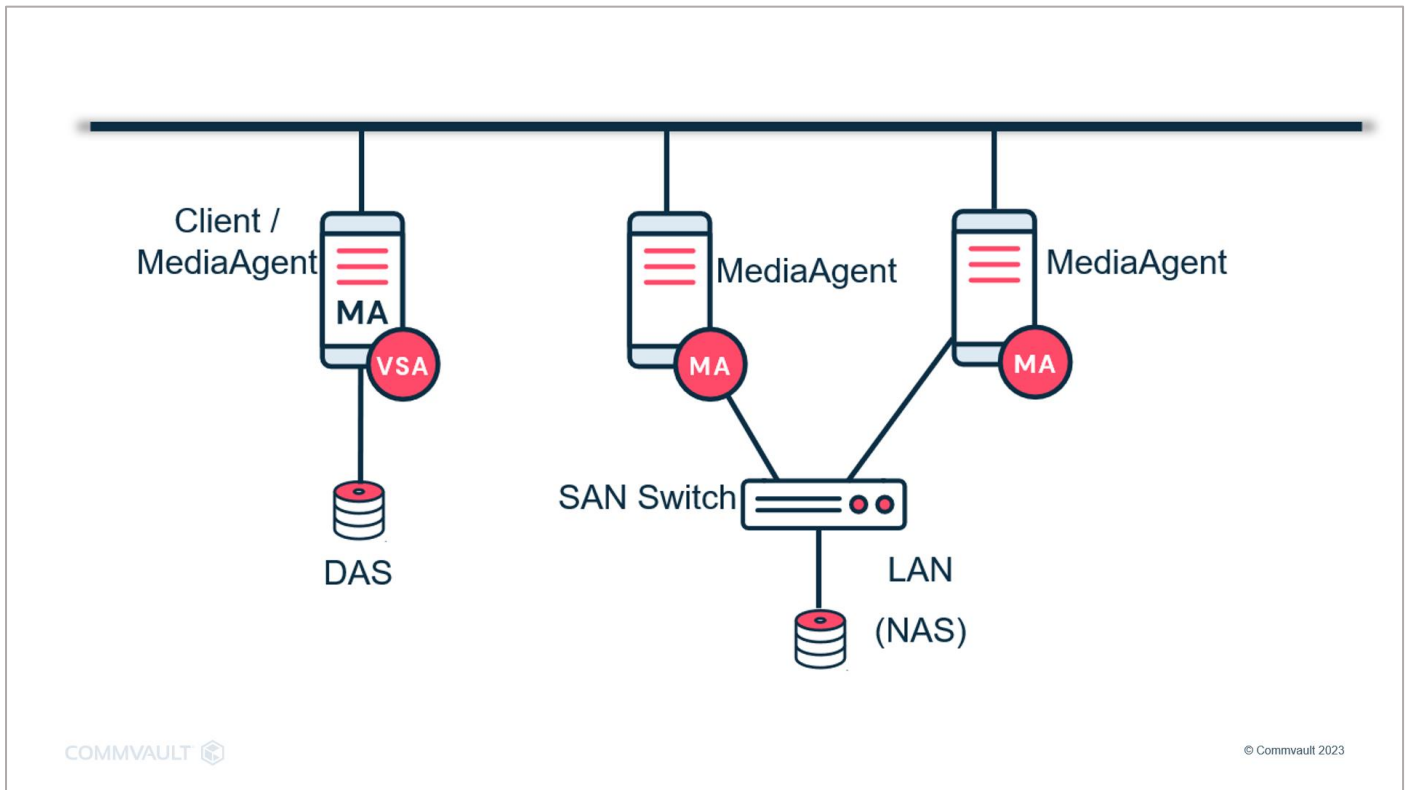
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Commvault software logically addresses storage systems to allow virtually any storage target to be used. The three primary target types are disk, cloud, and tape.

Disk Storage

Disk Storage Design



A Commvault disk storage is a logical container which is used to define one or more paths to storage called backup locations (mount paths). These paths are defined explicitly to the location of the storage as a drive letter or a UNC path. Within each mount path writers are allocated that define the number of concurrent streams for the mount path.

Disk Storage Types

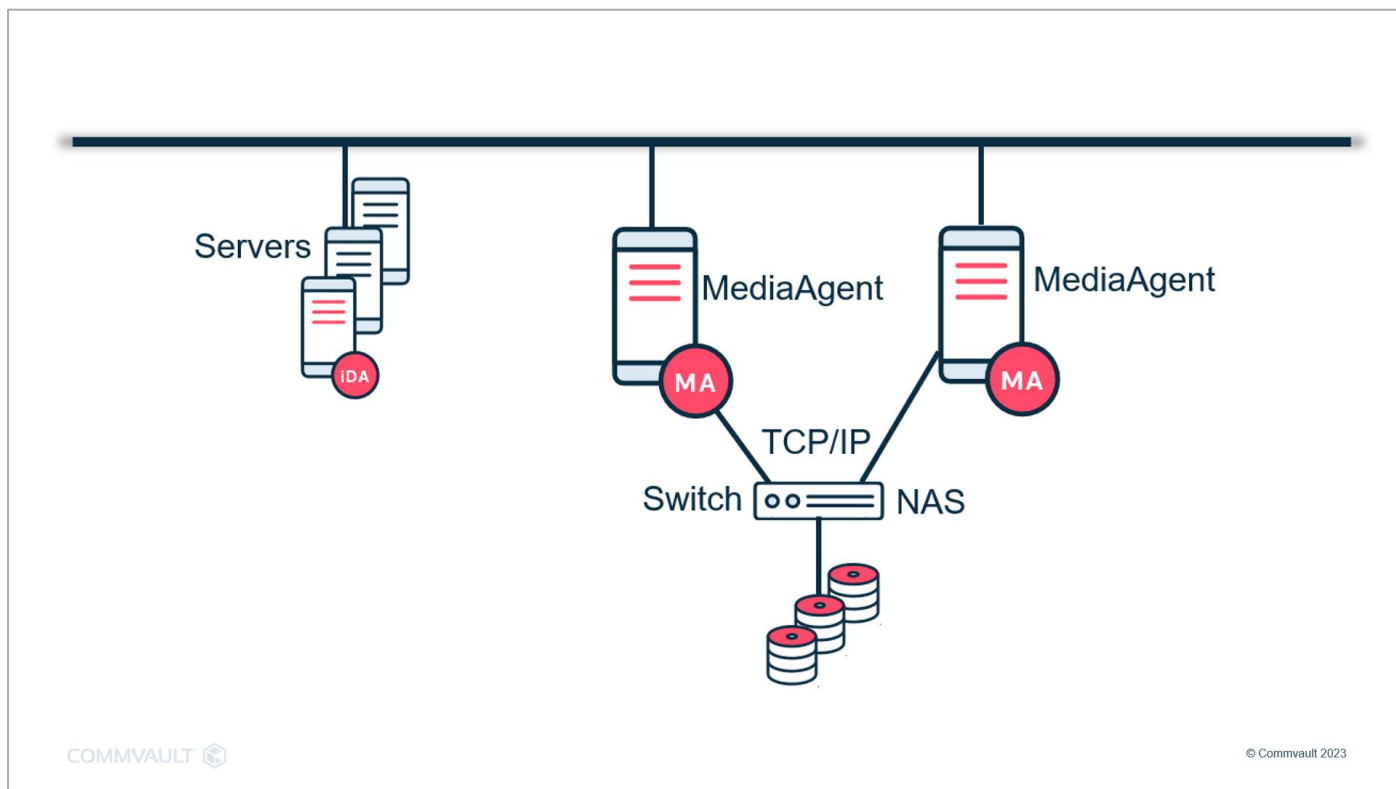
- **Dedicated** disk storage is one or more backup locations that are configured as shared disk devices. The shared disk device in a dedicated disk storage has only one primary sharing folder.
- **Shared** disk storage is a target with more than one primary sharing folder configured on a shared disk device. This enables other MediaAgents access to the same shared volume resource. Common Internet File System (CIFS) protocol is used to manage multiple MediaAgent access to the same directory. For UNIX hosted MediaAgents, Network File System (NFS) protocol can be used. NFS shared disks appear to the MediaAgent as local drives.

Disk storage can be configured as network-attached storage (NAS), storage area network (SAN), or direct-attached storage (DAS).

Disk Storage Best Practices

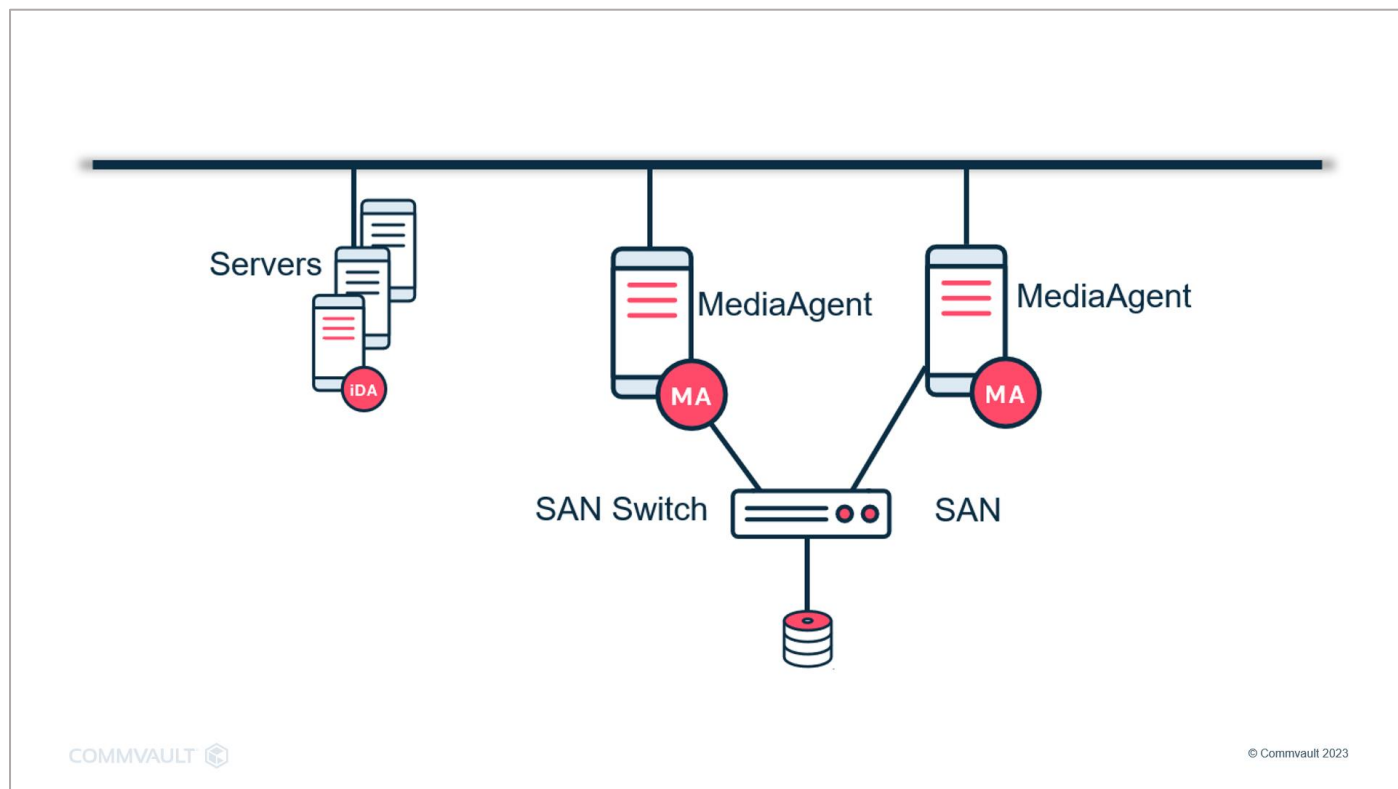
- If using DAS or SAN, format mount paths (backup locations) using a 64KB block size.
- If using DAS or SAN, create multiple backup locations (mount paths). For instance, if there are 10 mount paths, and there is a maintenance job, such as a defrag job running on one, the mount path can be set to read-only, leaving 90% of the storage available for backup jobs.
- Share the disk storage if required.
- For best results, enable Commvault deduplication.

Network Attached Storage (NAS)



Network-attached storage provides the best connection method for resiliency since the storage is accessed directly through the NAS device. This means that by using a Common Internet File System (CIFS) or a Network File System (NFS), Universal Naming Convention (UNC) paths can be configured to read and write directly to storage. In this case, the target can be configured as a shared disk storage, where all MediaAgents can see stored data for data protection and recovery operations.

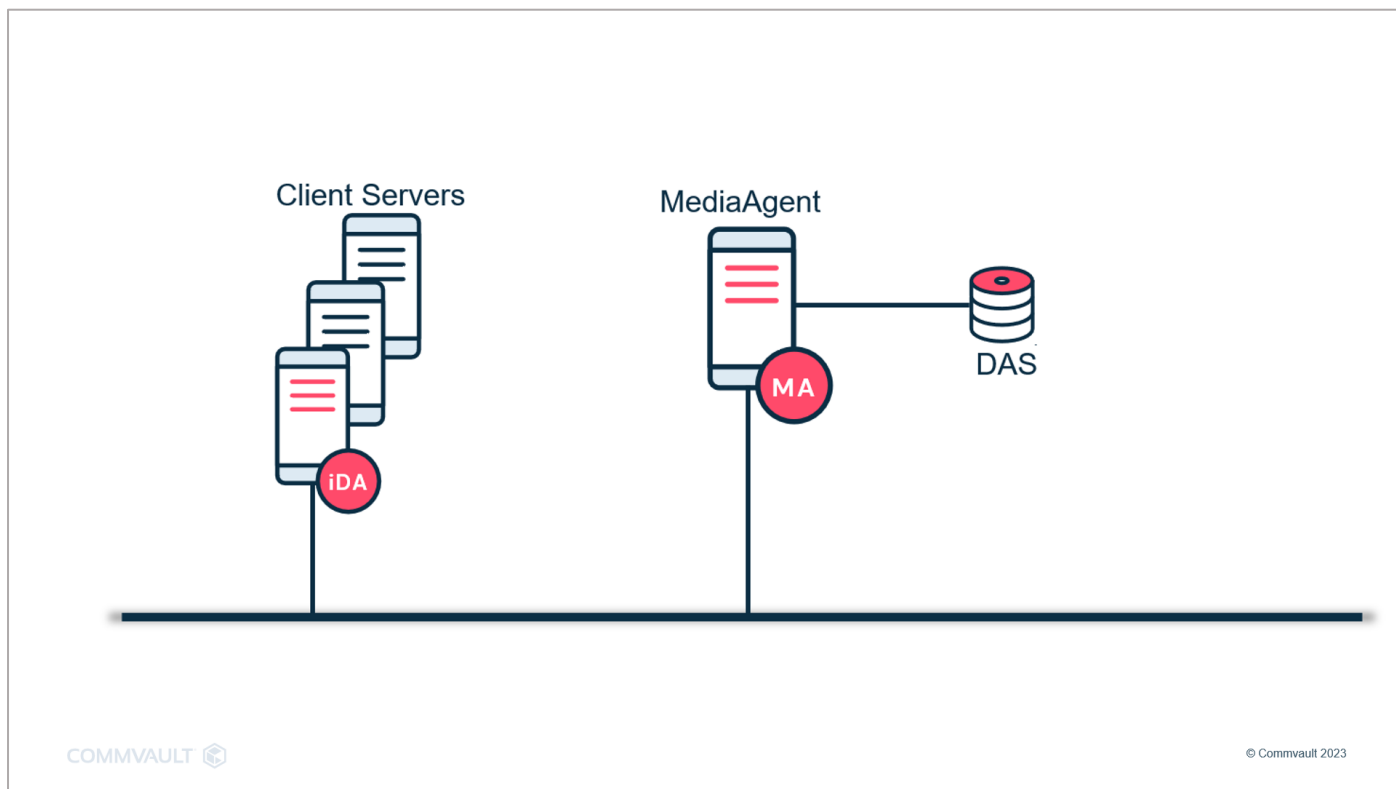
Storage Area Network (SAN)



Storage area networks or SANs are common in data centers. SAN storage can be zoned and presented to MediaAgents using either Fibre Channel or iSCSI. The zoned storage is presented directly to the MediaAgent providing read/write access to the disks.

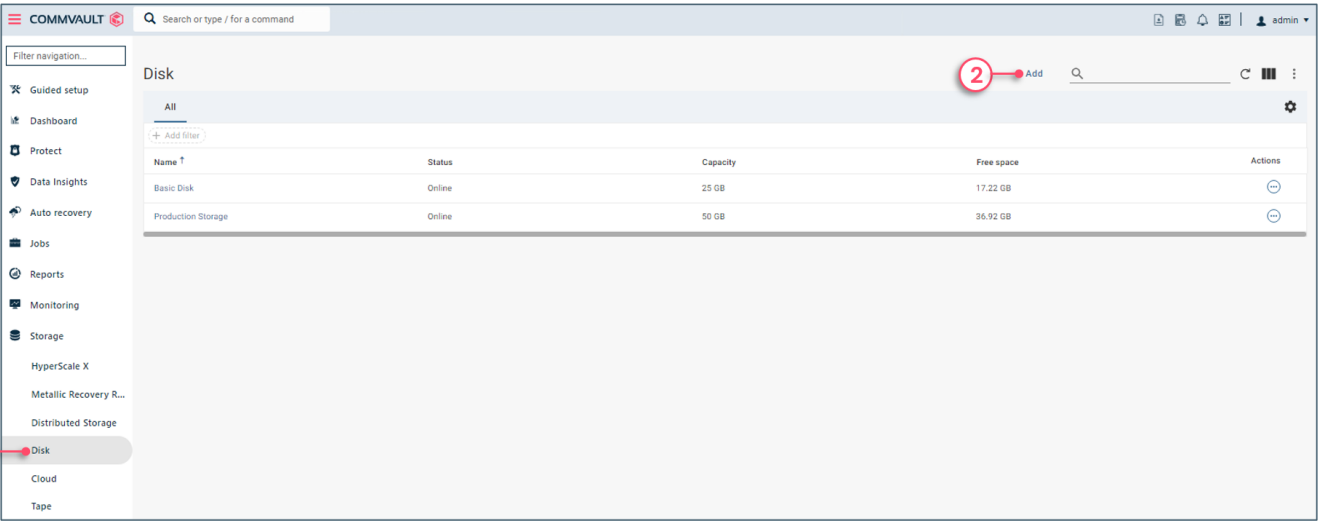
When using SAN storage, each building block should use a dedicated MediaAgent, DDB and Commvault disk storage. Although the back-end disk storage in the SAN can reside on the same disk array, it should be configured in the Commvault software as two separate Commvault disk storage where logical unit numbers (LUNs) are presented as mount paths (backup locations) in dedicated targets for specific MediaAgents.

Direct Attached Storage (DAS)



Direct-attached storage physically attaches the disk storage to the MediaAgent. Each building block is completely self-contained. This provides high performance but does not provide resiliency. If the MediaAgent controlling the building block fails, data stored in the disk storage cannot be recovered until the MediaAgent is repaired or replaced. All data in the disk storage is still completely indexed and recoverable, even if the index directory is lost. Once the MediaAgent is rebuilt, data from the disk storage can be restored.

Add Local or Network Disk as Storage



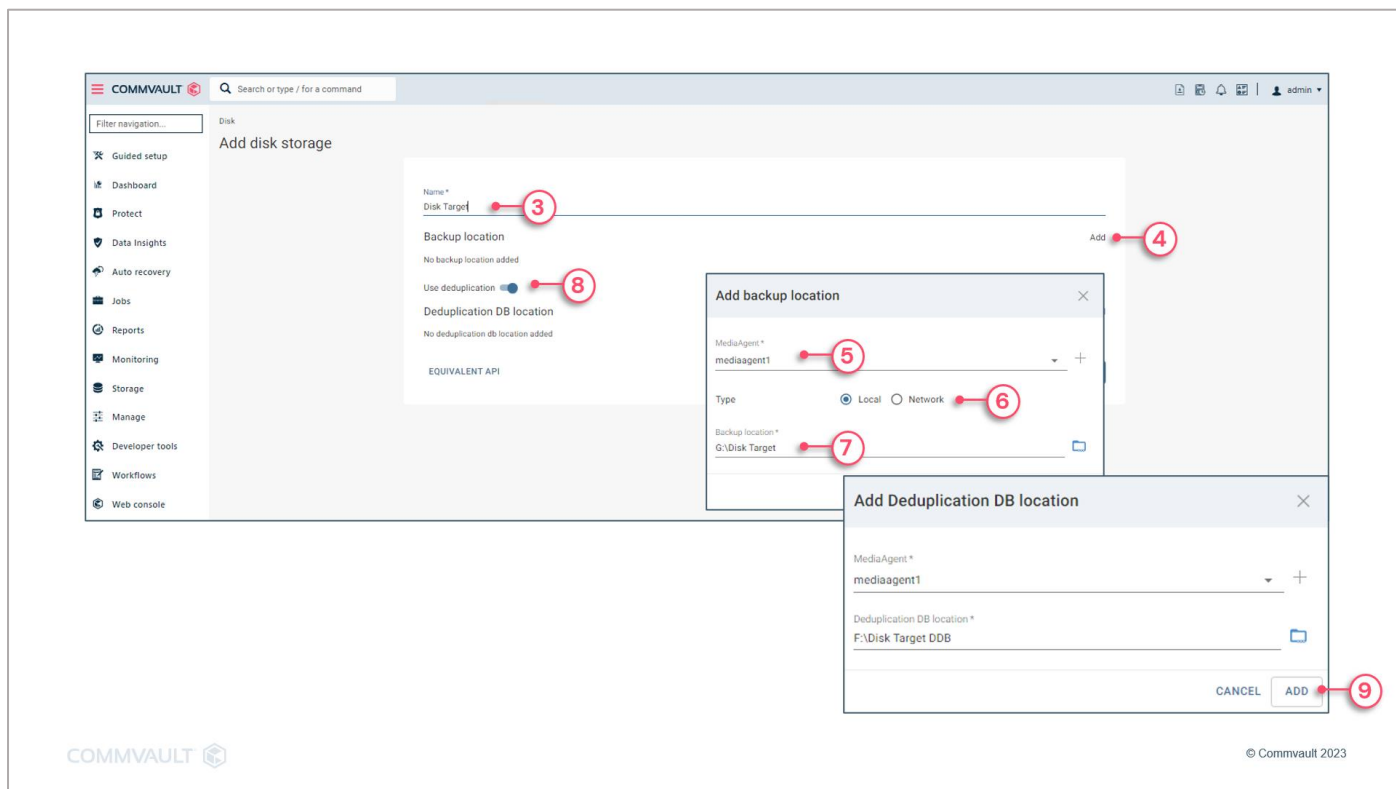
The screenshot shows the Commvault web interface. On the left, the navigation menu is expanded, and 'Disk' is selected under the 'Storage' category. In the main content area, the 'Disk' page is displayed. At the top right of this page is an 'Add' button. Below the 'Add' button is a table showing the configured disk storage.

Name ↑	Status	Capacity	Free space	Actions
Basic Disk	Online	25 GB	17.22 GB	⋮
Production Storage	Online	50 GB	36.92 GB	⋮

Before adding a Commvault disk storage, the operating system must have access to the storage. For SAN or DAS storage, volumes must be created and formatted in the operating system. For NAS, access the storage CIFS shares using the credentials.

1. Select **Storage > Disk** to view the configured disk storage
2. Click **Add**

Add Local or Network Disk as Storage ...cont'd



3. Enter a name
4. Click **Add**
5. Select the MediaAgent used to reach the target storage
6. Select the type
 - **Local** for DAS and SAN targets
 - **Network** path for NAS targets
7. Browse to the location of the first backup location (mount path)
8. Toggle **Use deduplication** on/off to enable/disable deduplication for the disk storage
9. Click **ADD** to select the Deduplication DB location

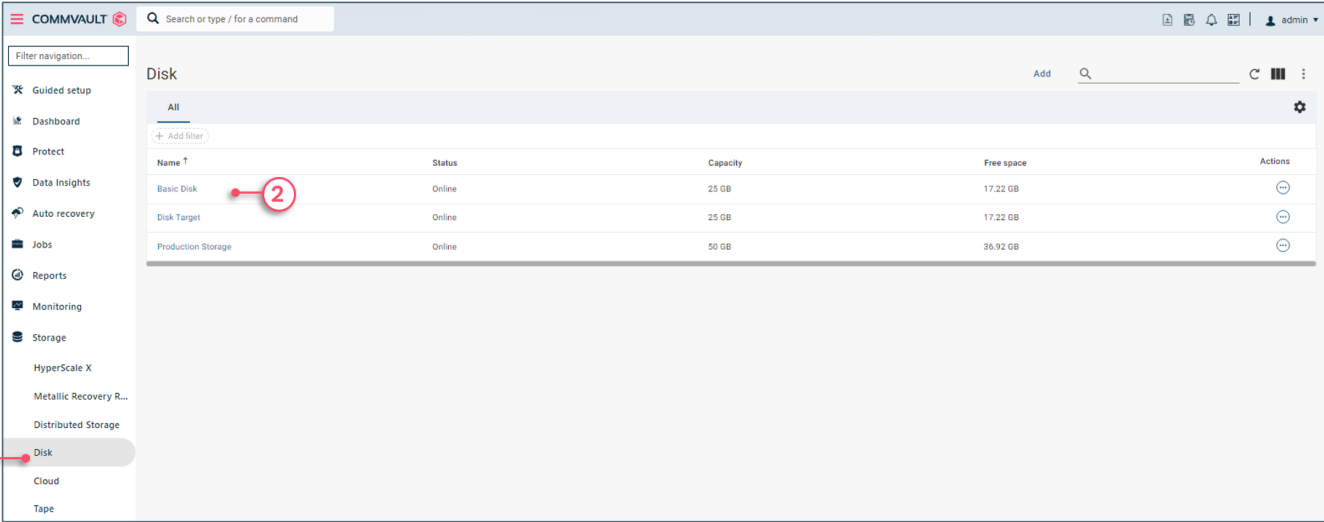
Add Local or Network Disk as Storage ...cont'd

The screenshot shows the 'Add disk storage' configuration window in the Commvault interface. The window has a sidebar on the left with navigation options like 'Guided setup', 'Dashboard', 'Protect', 'Data Insights', 'Auto recovery', 'Jobs', 'Reports', 'Monitoring', 'Storage', 'Manage', 'Developer tools', 'Workflows', and 'Web console'. The main area is titled 'Add disk storage' and contains the following fields:

- Name ***: A text input field with the value 'Disk Target'.
- Backup location**: A table with two columns: 'MediaAgent ↑' and 'Backup location'. It contains one row with 'mediaagent1' and 'G:\Disk Target'. There is an 'Add' button to the right of the table.
- Use deduplication**: A toggle switch that is currently turned on.
- Deduplication DB location**: A table with two columns: 'MediaAgent ↑' and 'Deduplication DB location'. It contains one row with 'mediaagent1' and 'F:\Disk Target DDB'. There is an 'Add' button to the right of the table.
- EQUIVALENT API**: A text input field.
- CANCEL** and **SAVE** buttons: Located at the bottom right of the window. A red circle with the number 10 points to the 'SAVE' button.

10. Click **SAVE** to complete the disk storage configuration

Add Backup Locations to Disk Storage



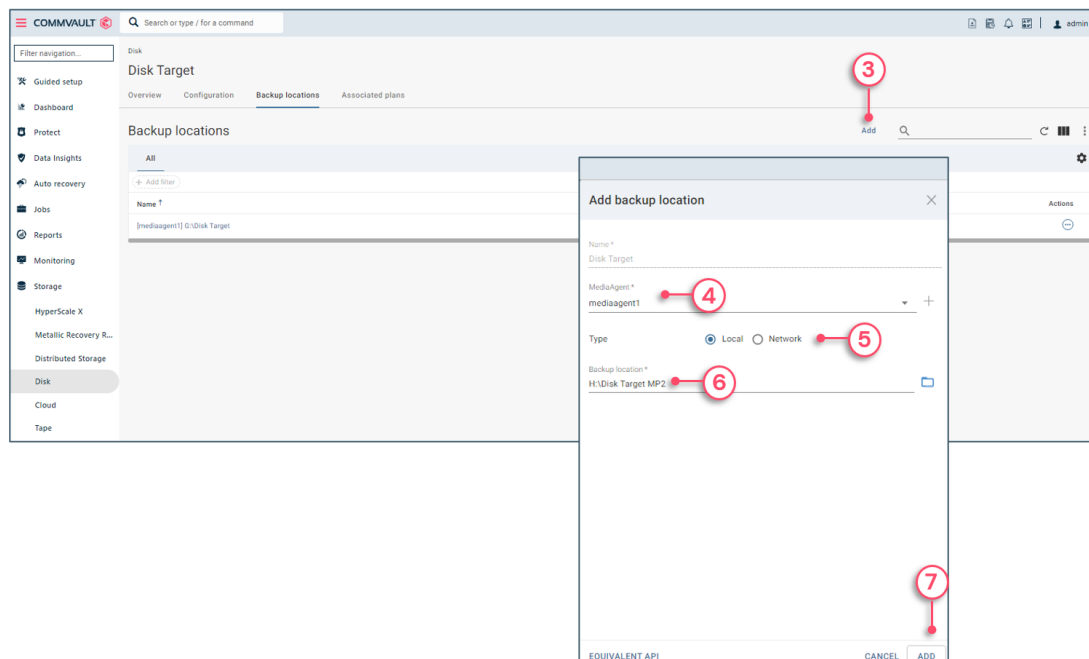
Name ↑	Status	Capacity	Free space	Actions
Basic Disk	Online	25 GB	17.22 GB	⋮
Disk Target	Online	25 GB	17.22 GB	⋮
Production Storage	Online	50 GB	36.92 GB	⋮

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Creating the disk storage also defines the first backup location (mount path). Any additional backup location will need to be defined.

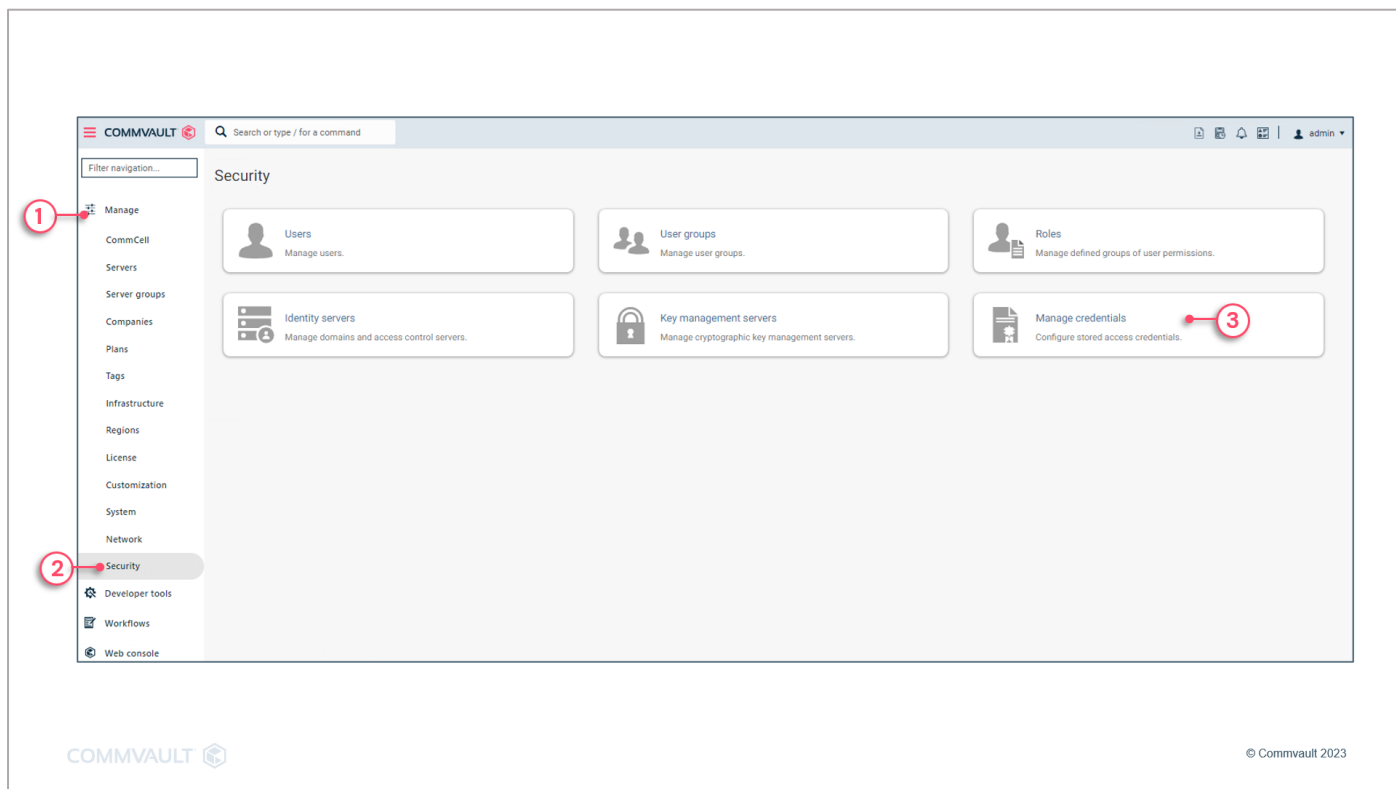
1. Select **Storage > Disk** to view the configured disk storage
2. Click the name of the disk storage for which you want to add a mount path

Add Backup Locations to Disk Storage ...cont'd



3. In the *Backup Locations* tab, click **Add** to add a new mount path
4. Select the MediaAgent that has access to the mount path
5. Select a local or network path
6. Browse for the mount path location
7. Click **ADD** to add the mount path

Share Disk Storage – Credentials



If the Commvault storage is using NAS, it can be shared with other MediaAgents. Sharing storage is achieved by sharing its backup location. Apply the sharing procedure on each backup location (mount path).

1. Select **Manage**
2. Select **Security**
3. Click the **Manage credentials** tile

Share Disk Storage – Credentials ...cont'd

The image displays two screenshots of the Commvault Security console interface. The top screenshot shows the 'Manage credentials' window with a table of existing credentials. The bottom screenshot shows the 'Add credential' form with numbered callouts 4 through 9 indicating the steps to add a new credential.

Manage credentials window (top):

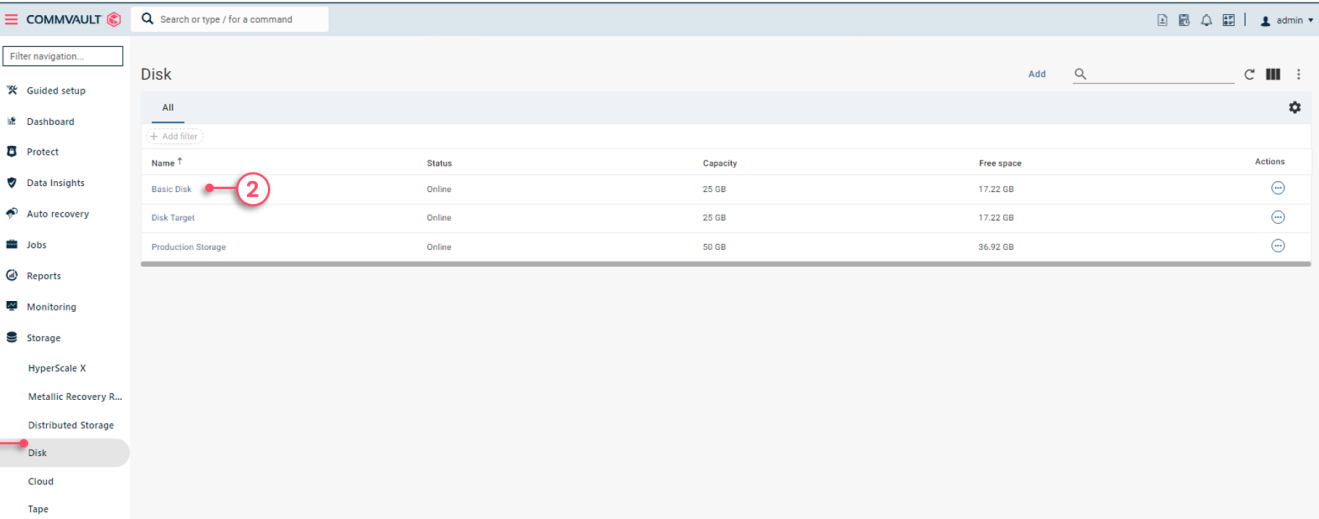
- Search bar: Search or type / for a command
- Filter navigation: Filter navigation...
- Security / Manage credentials
- Table columns: Credential name, Type, Description, Actions
- Table data: Storage Admin Access, S3 Compatible Storage, No description
- Callout 4: Points to the 'Add' button in the top right corner.

Add credential window (bottom):

- Search bar: Search or type / for a command
- Filter navigation: Filter navigation...
- Security / Manage credentials
- Form fields: Account type, Credential vault, Credential name, User account, Password, Description
- Callouts: 5 (Account type), 6 (Credential vault), 7 (Credential name), 8 (User account), 9 (Password)
- Buttons: CANCEL, SAVE

4. In the *Manage Credentials* window, click **Add**
5. Enter a security account from the account type selected
6. Select the credential vault to be used
7. Enter a name for the credential
8. Enter the user account and password for the credential
9. Click **SAVE**

Share Disk Storage

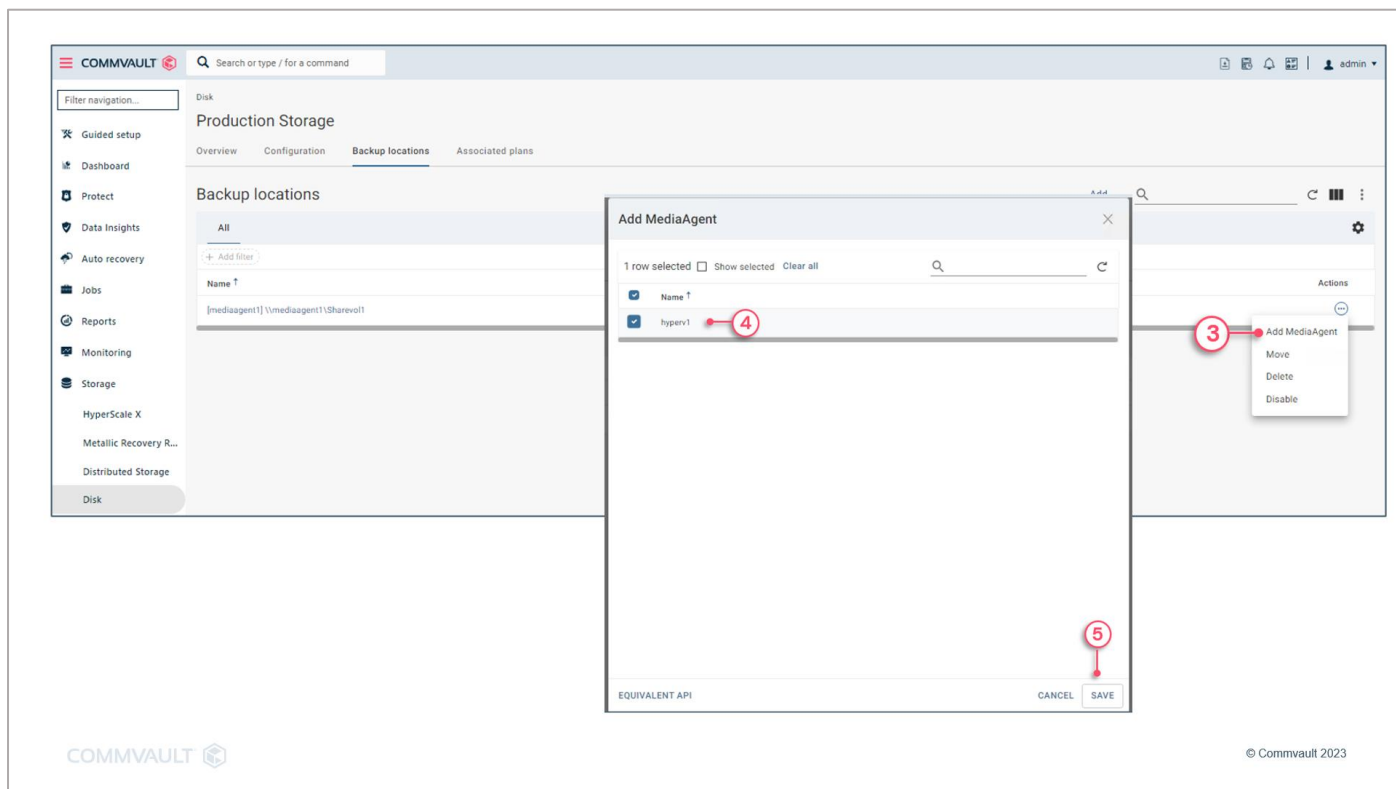


The screenshot shows the Commvault web interface. On the left, the navigation menu is expanded, and 'Disk' is selected under the 'Storage' category. In the main content area, a table lists the configured disk storage. The first row, 'Basic Disk', is highlighted. The table has the following columns: Name, Status, Capacity, Free space, and Actions.

Name ↑	Status	Capacity	Free space	Actions
Basic Disk	Online	25 GB	17.22 GB	⋮
Disk Target	Online	25 GB	17.22 GB	⋮
Production Storage	Online	50 GB	36.92 GB	⋮

1. Select **Storage > Disk** to view the configured disk storage
2. Click the name of the disk storage for which you want to share the backup locations (mount paths)

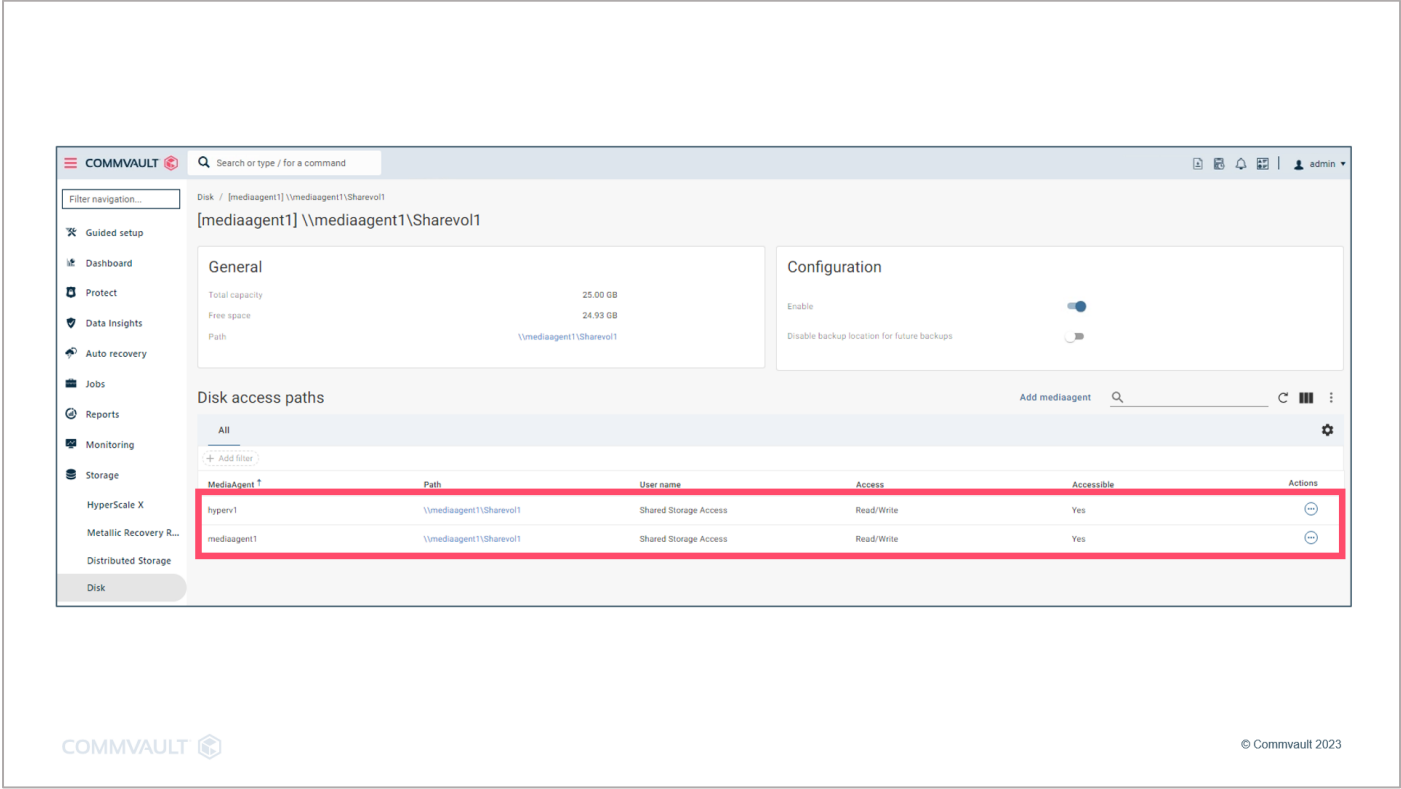
Share Disk Storage ...cont'd



This view displays the paths accessing the disks. In this example, only one MediaAgent has access to the disk

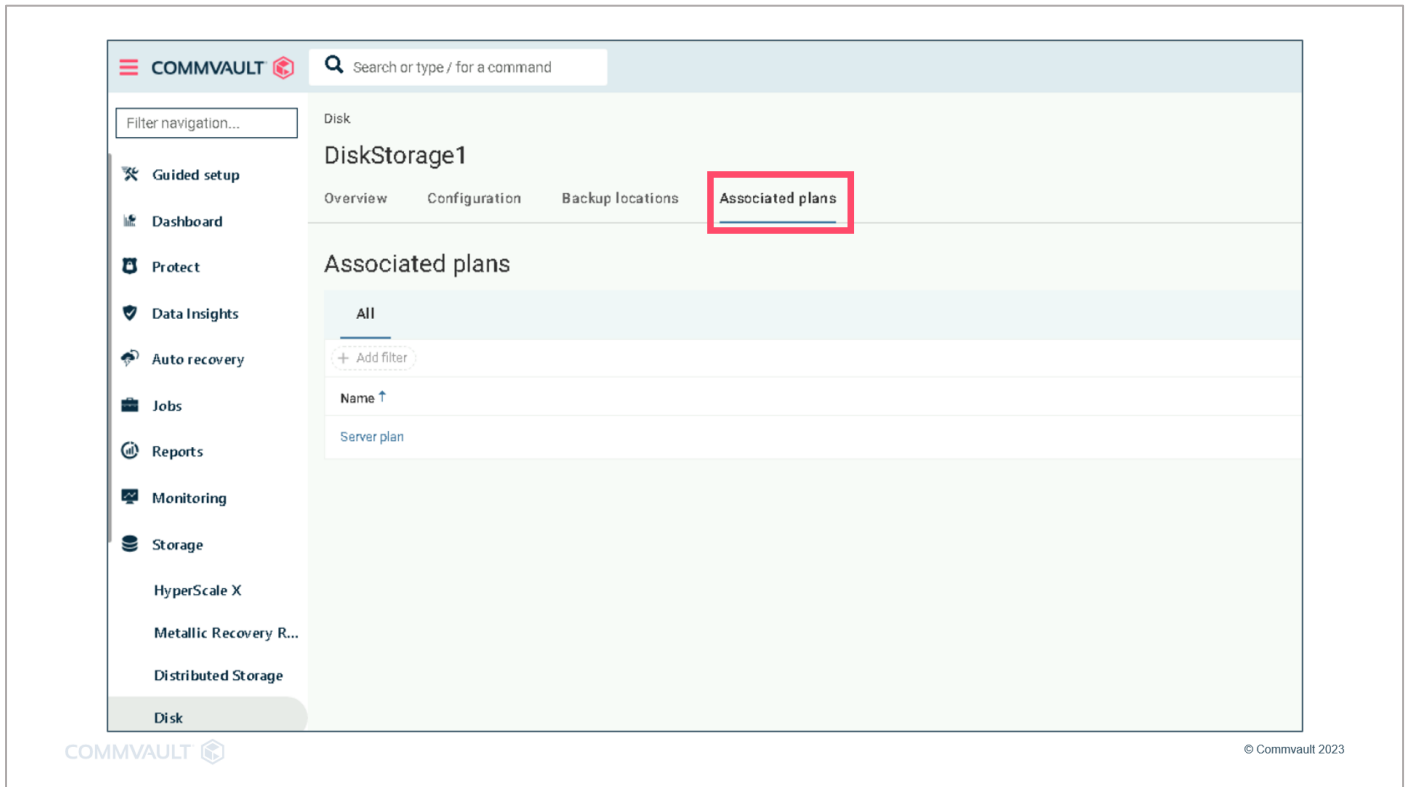
3. Click the **Actions** icon and then click **Add MediaAgent**
4. Select the MediaAgent with which the backup location (mount path) will be shared
5. Click **SAVE** to add the MediaAgent

Share Disk Storage ...cont'd



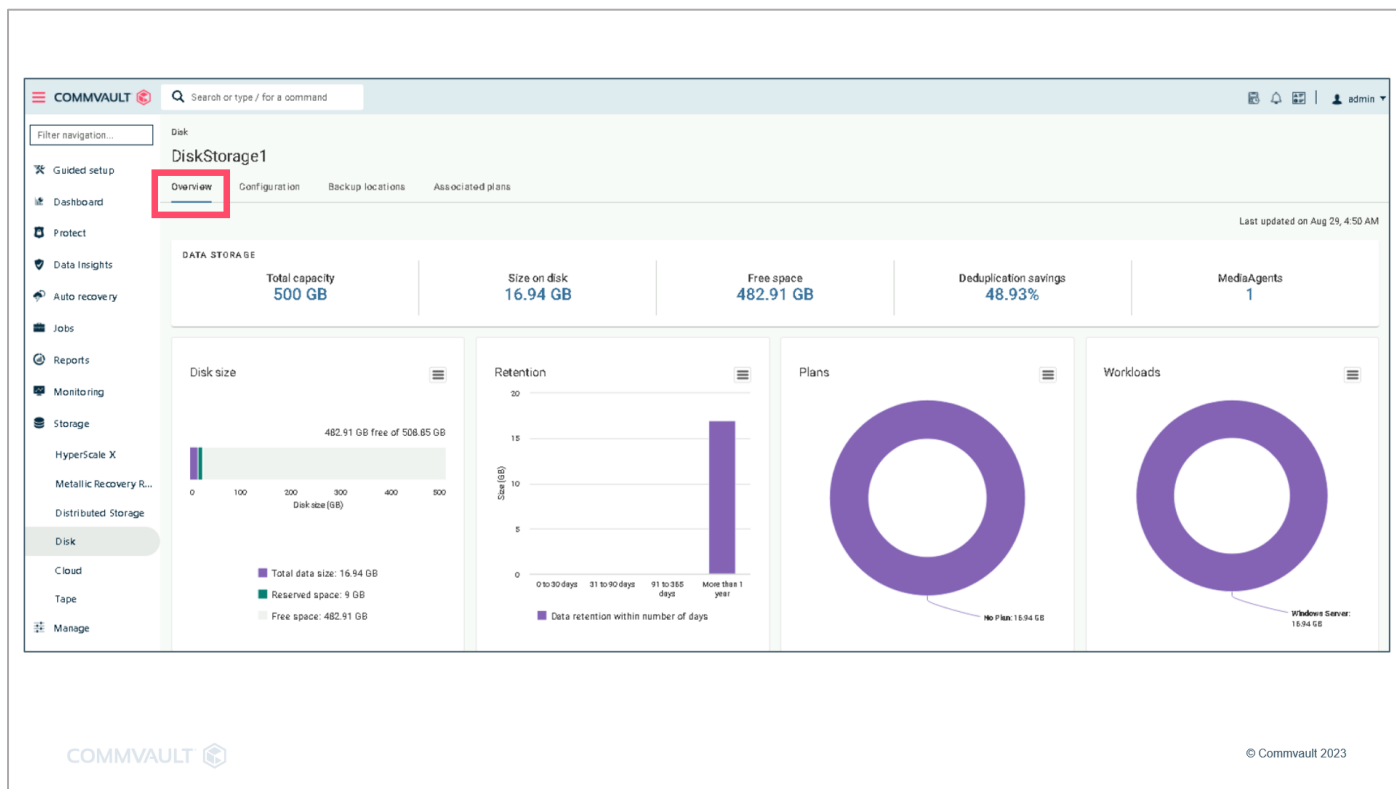
The second access path is added to the list. Repeat the same process to all other backup locations.

Disk Library Association Storage Plans



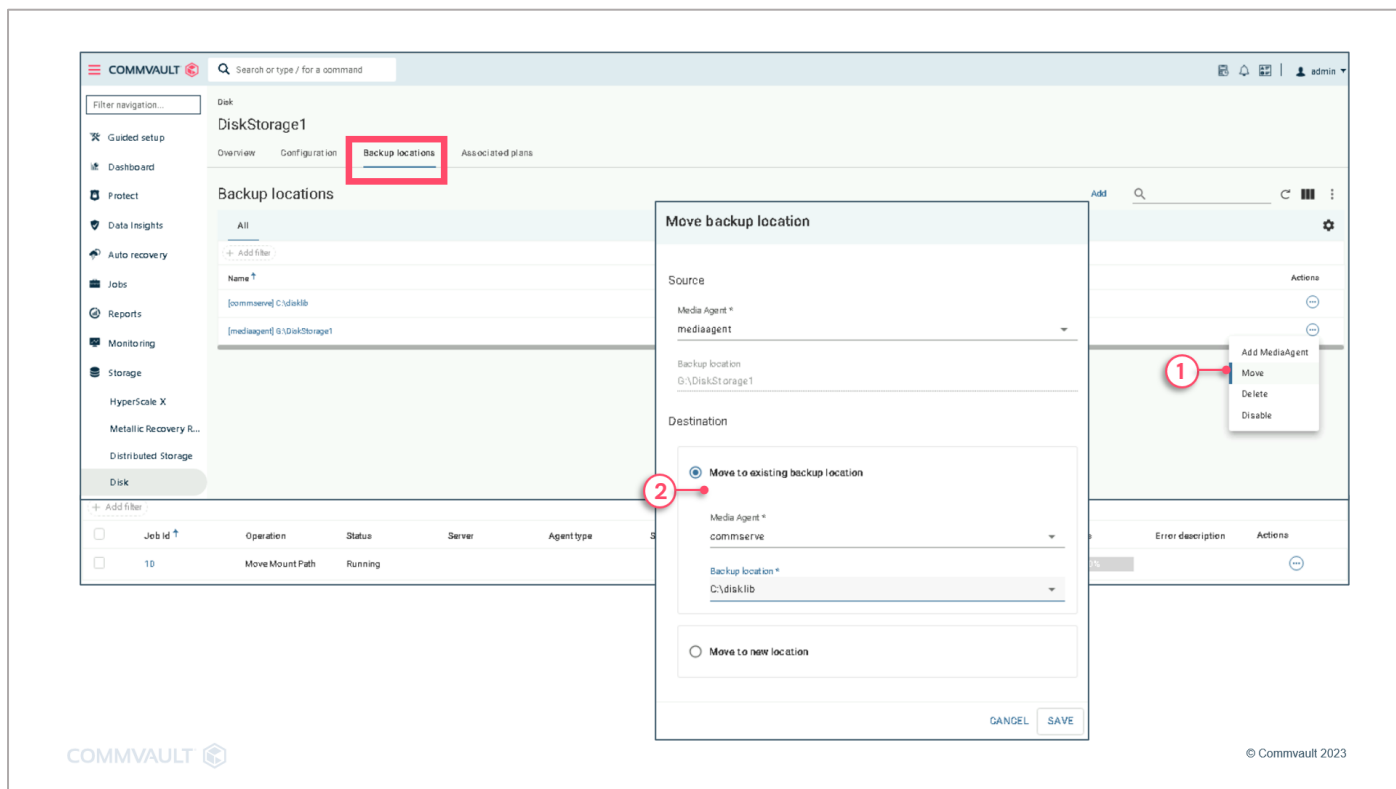
The *Associated Plans* tab of a disk library displays all storage plans that have a copy sending data to the disk library. This is useful information when you plan on decommissioning a disk library or taking it offline for maintenance.

Disk Library Usage



The disk library *Overview* tab displays the space usage, deduplication space saving and data aging pruning information. Keeping an eye on this tab helps in avoiding filled library situations and helps in planning the acquisition or replacement of hardware. Based on the average daily disk space consumption and the average space freed up by data aging, the system can estimate a possible date on which the library is expected to be filled.

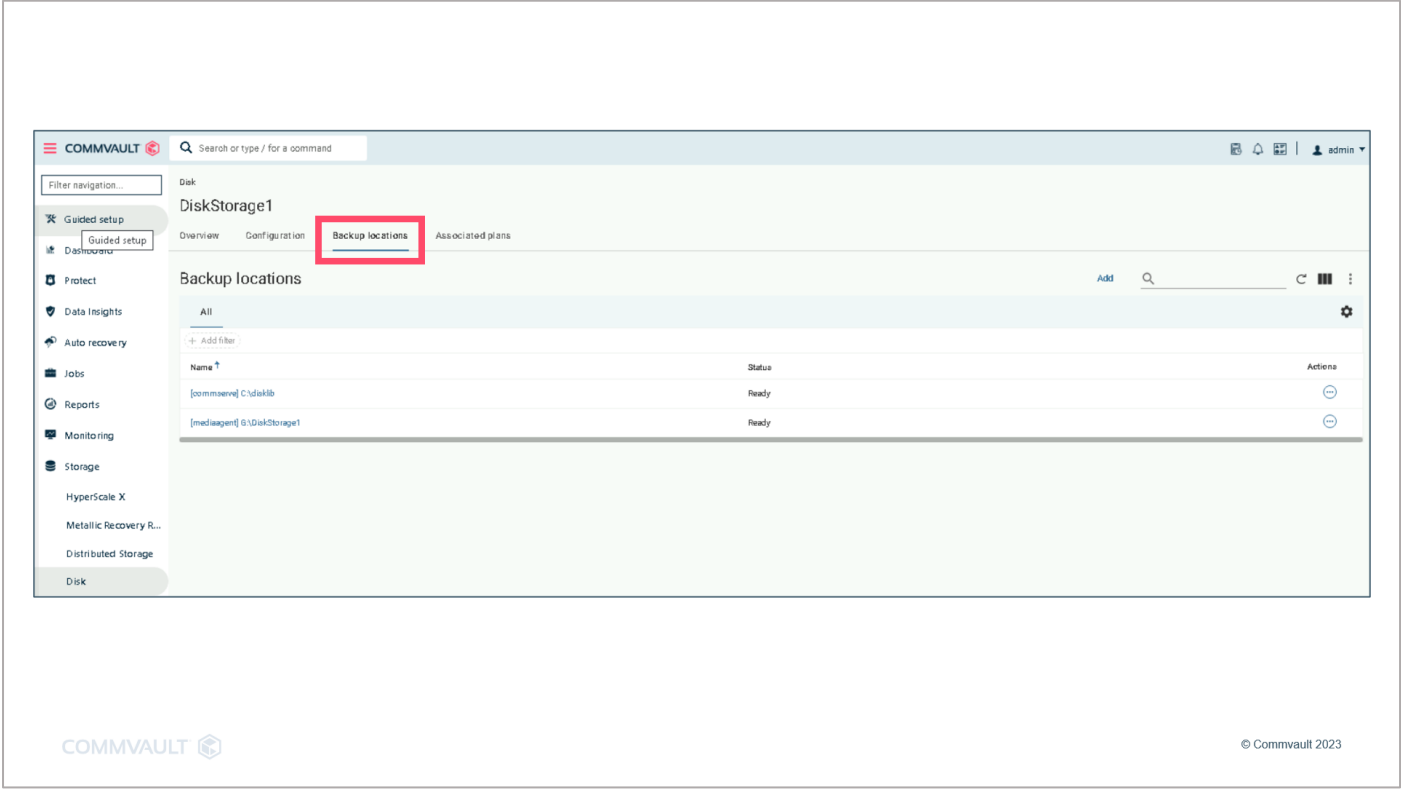
Move a Disk Library Mount Path



A disk library mount path can be moved to a different location using a 'Move Data Path' operation. This fully automated functionality is used when a disk library mount path is lacking space and needs to be moved to a bigger storage volume, or when an older storage unit is replaced. The new storage must first be presented to the MediaAgent operating system.

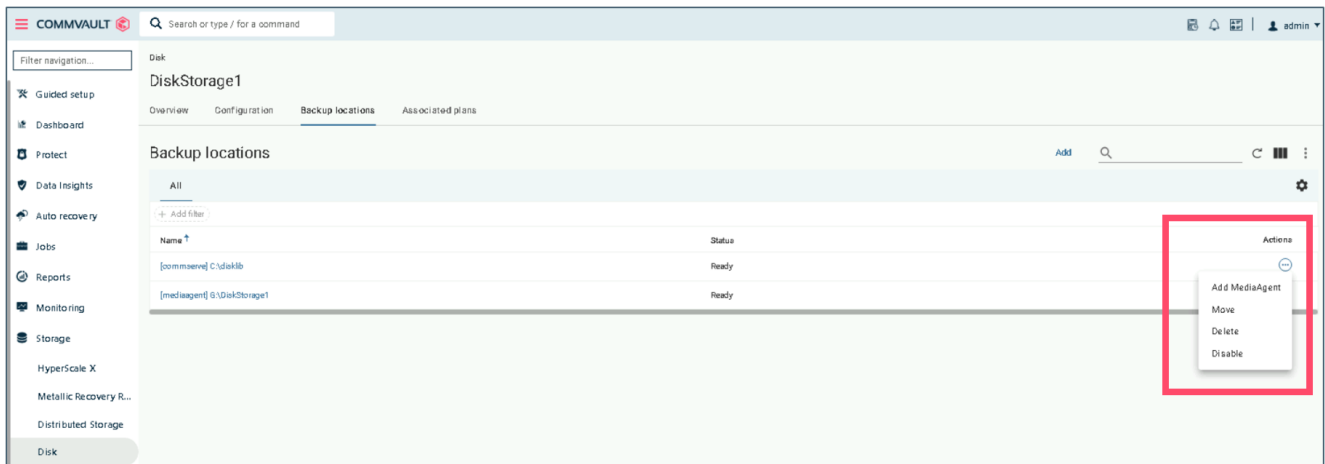
1. From the backup location you wish to move, click the **Actions icon** and then click **Move**
2. Choose to move to an existing backup location or to a new one and click **SAVE**

Disk Library Mount Path Status



The *Backup locations* tab will show the status for all configured mount paths.

Disable Disk Library or Mount Path



By clicking the *Actions* icon next to a mount path, you are given the option to disable a mount path.

Set a Mount Path as Read Only

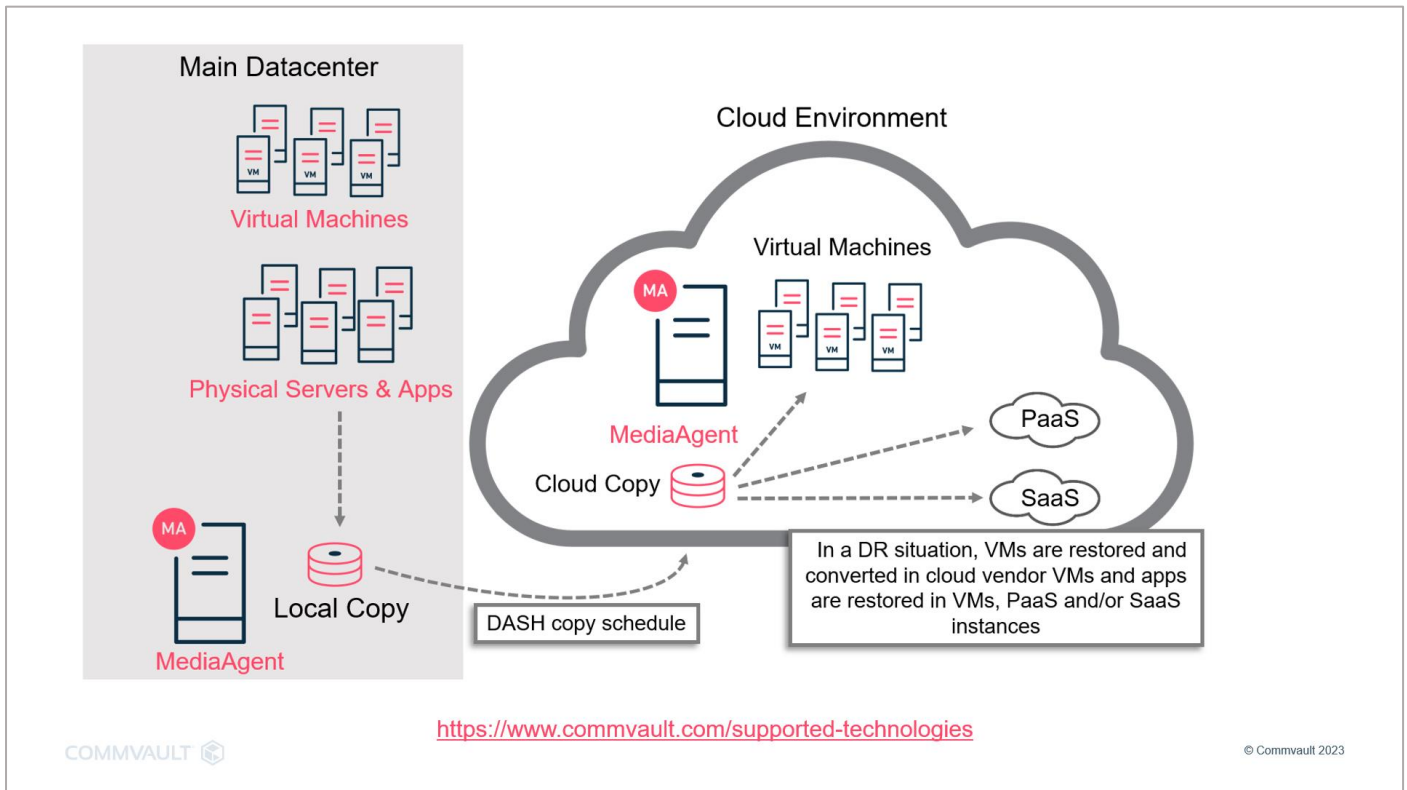
The screenshot displays the Commvault console interface. On the left is a navigation sidebar with options like Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, Storage, HyperScale X, Metallic Recovery R..., Distributed Storage, and Disk. The main area shows the 'Disk' section for the path '[commserve] C:\disklib'. It has two tabs: 'General' and 'Configuration'. The 'Configuration' tab is highlighted with a red rectangle and contains two toggle switches: 'Enable' and 'Disable backup location for future backups', both of which are currently turned off. Below the tabs is a 'Disk access paths' section with a table listing access paths.

MediaAgent	Path	User name	Access	Accessible	Actions
commserve	C:\disklib		Read/Write	Yes	⋮
mediaagent	via network		Read	Yes	⋮

In the *Configuration* pane for a selected mount path, you can set the mount path to read-only by enabling the *Disable backup location for future backups* toggle switch.

Cloud Storage

Cloud Storage Design



Cloud storage is managed through the Commvault Command Center. Commvault support cloud storage across many vendors, and also offer Commvault cloud solutions.

These are popular options for their availability and, in some cases, lower cost. Some of the drawbacks for using cloud storage for DR include limited bandwidth and data security.

Using advanced features such as Commvault deduplication can greatly reduce the bandwidth requirements. However, in a disaster situation where a significant amount of data must be restored, bandwidth can become a serious bottleneck.

Data transfers are achieved using secured channels (HTTPS) and are optionally encrypted to further secure the data sent to the cloud.

Cloud Storage Best Practices

- Properly plan and analyze if the cloud storage scenario meets the needs (e.g., restoring an entire data center)
- If the link is shared with users, consider throttling Commvault bandwidth usage during business hours
- If the MediaAgent does not have direct access to the Internet, define the access nodes settings in the cloud storage access nodes configuration page
- If the cloud storage is accessed through a high-speed internet link (1GB or higher), consider tuning the connection

For more information, refer to the Commvault Online Documentation. Go to <https://www.commvault.com/supported-technologies> to view the current list of supported cloud vendors.

Add Cloud Storage

The screenshot shows the Commvault web interface. On the left, a navigation sidebar lists various sections: Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, Storage, HyperScale X, Metallic Recovery R..., Distributed Storage, Disk, and Cloud. A red circle with the number '1' points to the 'Cloud' option in this sidebar. The main content area is titled 'Cloud' and contains a table with the following columns: Name, Status, Capacity, Free space, and Actions. A red circle with the number '2' points to the 'DR Backups' entry in the 'Name' column. Above the table, there is a '+ Add filter' button and an 'Add' button, with a red circle and the number '3' pointing to the 'Add' button. The Commvault logo is in the bottom left, and the copyright notice '© Commvault 2023' is in the bottom right.

Name	Status	Capacity	Free space	Actions
DR Backups	Online	N/A	N/A	

1. Select **Storage > Cloud**
2. View the existing cloud storage, if any
3. Click **Add** to add cloud storage

Add Cloud Storage ...cont'd

The screenshot shows the 'Add cloud storage' configuration page in the Commvault interface. The page has a left sidebar with navigation options: Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, and Storage. The 'Storage' section is expanded, showing options like HyperScale X, Metallic Recovery R..., Distributed Storage, Disk, and Cloud. The main area is titled 'Add cloud storage' and contains a form with the following fields and callouts:

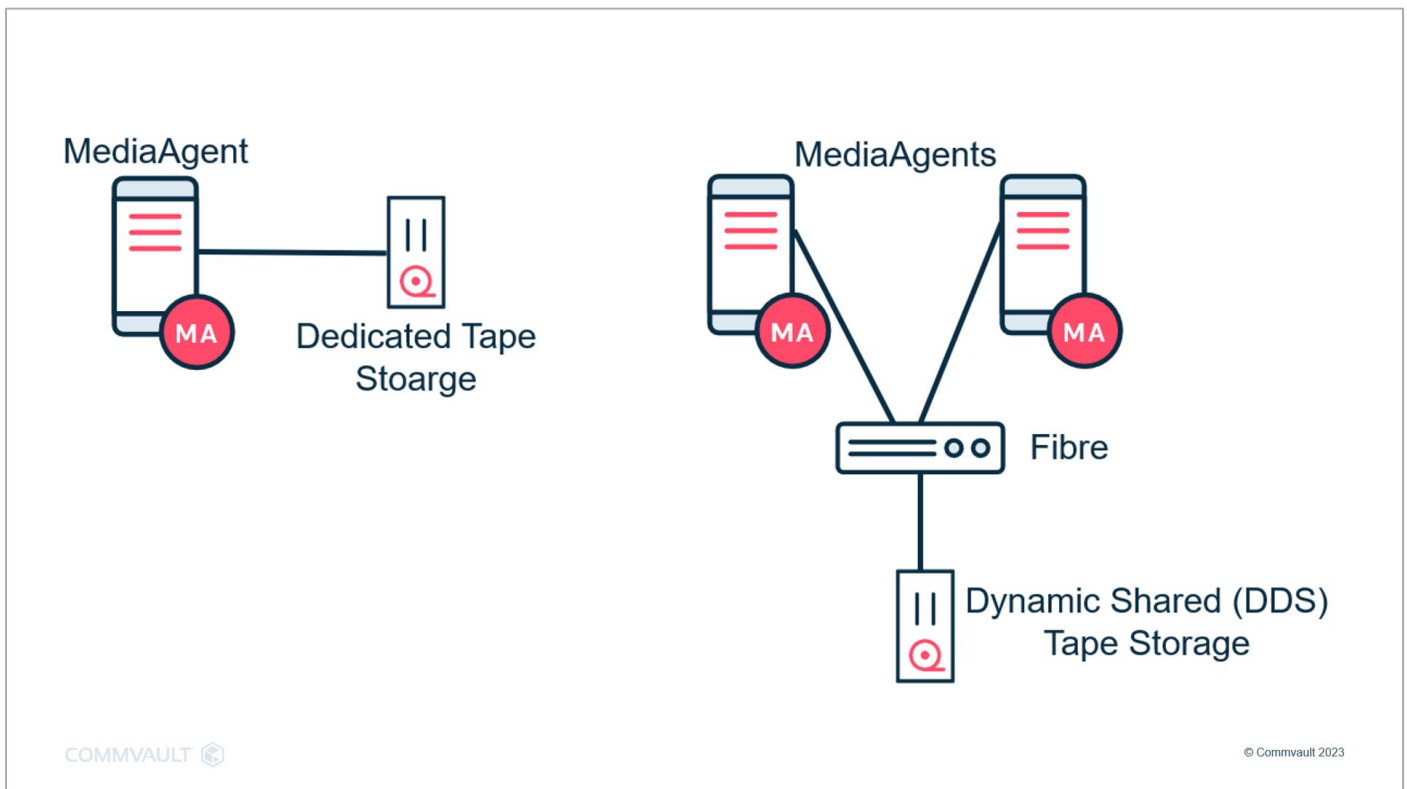
- 4**: Type (Dropdown menu, currently showing 'S3 Compatible Storage')
- 5**: Name * (Text input field, currently showing 'Private Cloud storage')
- 6**: MediaAgent * (Dropdown menu, currently showing 'mediaagent1')
- 7**: Service host * (Text input field, currently showing '1\CloudLib')
- 8**: Credentials * (Dropdown menu, currently showing 'Cloud Storage Access')
- 9**: A '+' icon next to the Credentials field to create a new credential set.
- 10**: Bucket * (Text input field, currently showing 'compliancecopies')
- 11**: A 'SAVE' button at the bottom right of the form.

At the bottom of the form, there is a checkbox for 'Use deduplication' and a link for 'EQUIVALENT API'. The bottom of the page features the Commvault logo and the copyright notice '© Commvault 2023'.

4. Enter a type
5. Choose the cloud provider
6. Select the MediaAgent with access to cloud storage
7. Choose the address of the cloud provider service
8. Select a saved cloud credential set from the list
9. If it does not exist, click + to create one
10. Enter the name of the bucket to store backup data and define a storage class if required by the vendor
11. Click **SAVE**

Tape Storage

Tape Storage Design



A tape storage is a target where media can be added, removed, and moved between multiple targets. The term “removable media” is used to specify various types of removable media supported by Commvault software, including tape and USB disk drives, which can be moved between MediaAgents for data protection and recovery operations.

Add Tape Storage

The image displays two screenshots of the Commvault software interface, illustrating the steps to add tape storage. The first screenshot shows the 'Tape' section in the left navigation pane (1) and the 'Add' button (3). The second screenshot shows the 'Add tape' dialog where a MediaAgent is selected (4), 'SCAN HARDWARE' is clicked (5), and 'SAVE' is clicked (6).

Table 1: Tape Storage List

Name	Status	Manufacturer	Model	Actions
CommVault Scalar i6000 3	Ready	CommVault	Scalar i6000	

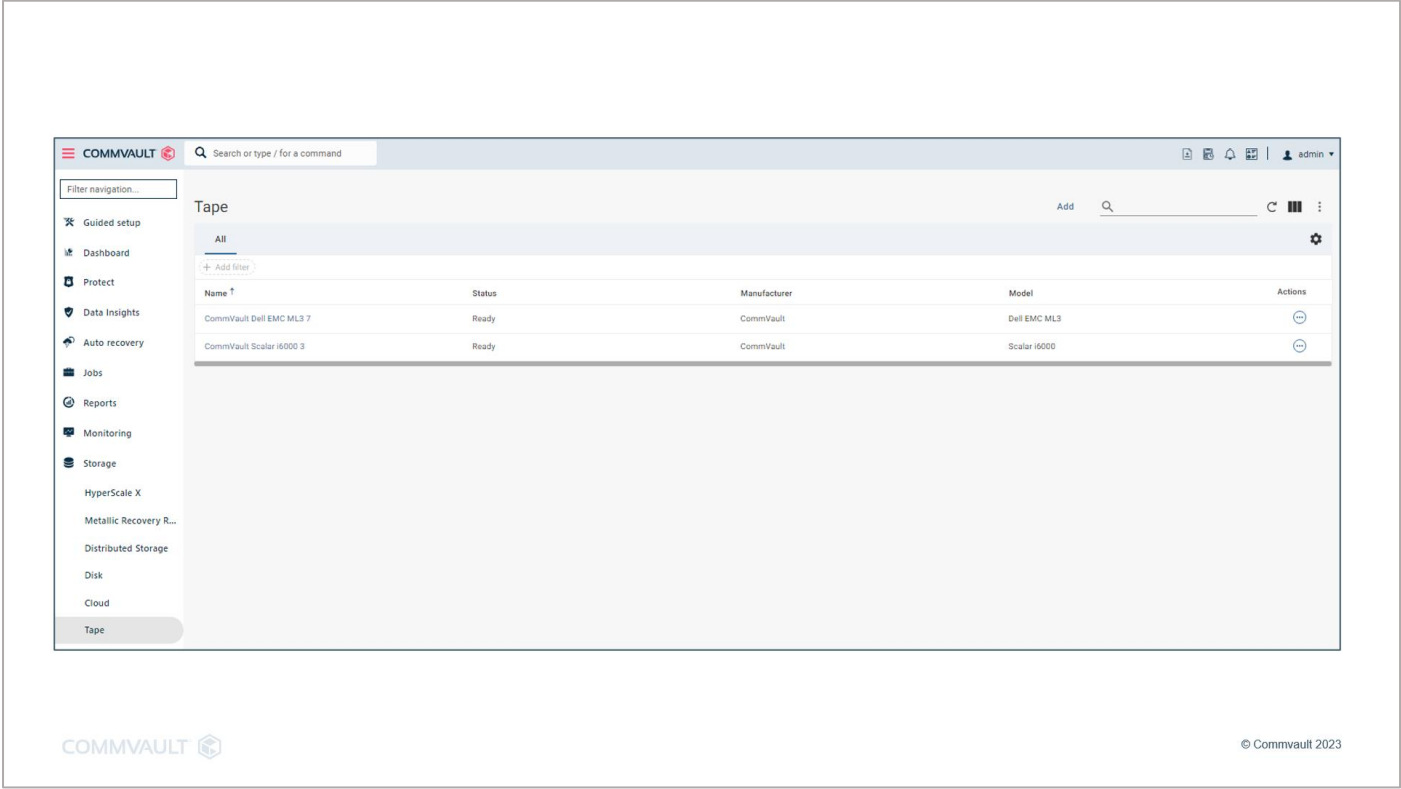
Table 2: MediaAgent Selection

MediaAgent	Status
mediaagent1	Configured
CommVault Dell EMC ML3	Not configured
CommVault VirtualDriveX2	Configured
CommVault VirtualDriveX2	Configured
CommVault VirtualDriveX2	Configured
CommVault VirtualDriveX2	Configured
CommVault VirtualDriveX2	Configured
CommVault VirtualDriveX2	Configured

Before you configure a tape storage in Commvault software, it first must be attached or zoned to be seen by the MediaAgent operating system. The device manager should detect one medium changer and one or many drives. It is recommended to install the vendor drivers specific to the target storage.

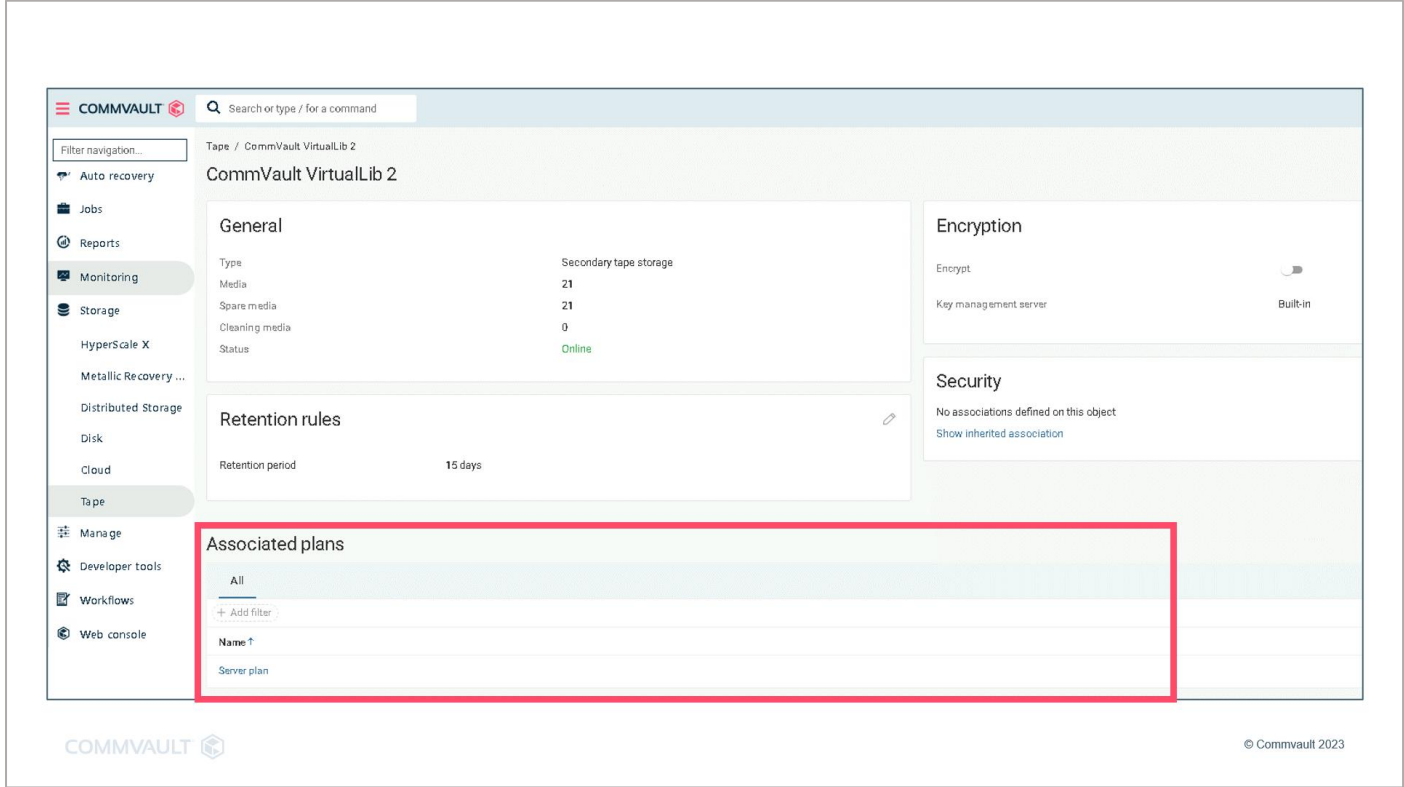
1. Select **Storage > Tape**
2. All existing tape storage are displayed in this view
3. Click **Add** to configure a new tape storage
4. Select the MediaAgent from the list
5. Click **SCAN HARDWARE** and select the unconfigured tape storage and its drives (recognizable by the red exclamation mark in a yellow circle)
6. Click **SAVE**

Add Tape Storage ...cont'd



The newly configured tape storage is displayed in the window.

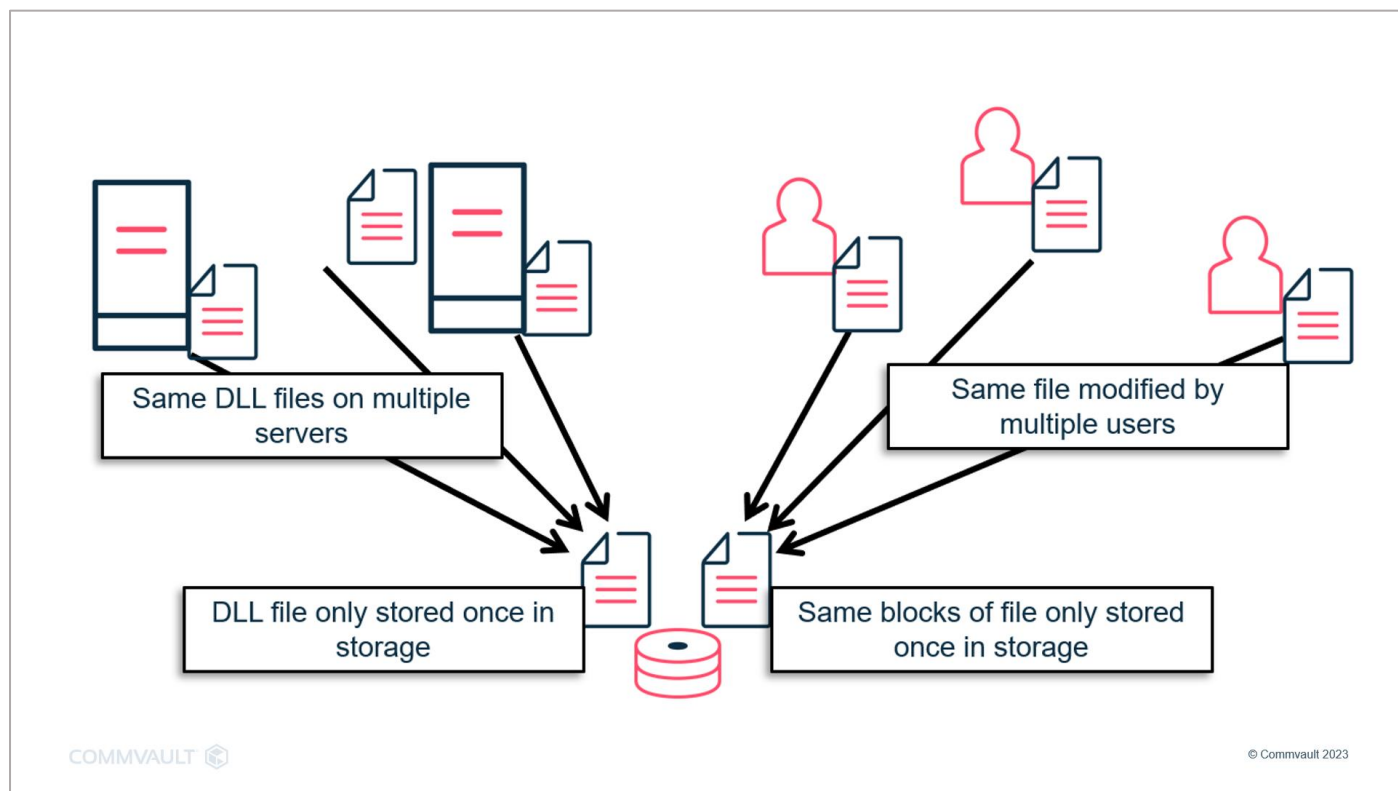
Tape Library Plan Association



The *Associated plans* section displays all plans that have a copy sending data to the tape library. This is useful information when you plan on decommissioning a library or taking it offline for maintenance.

Deduplication

Deduplication Overview



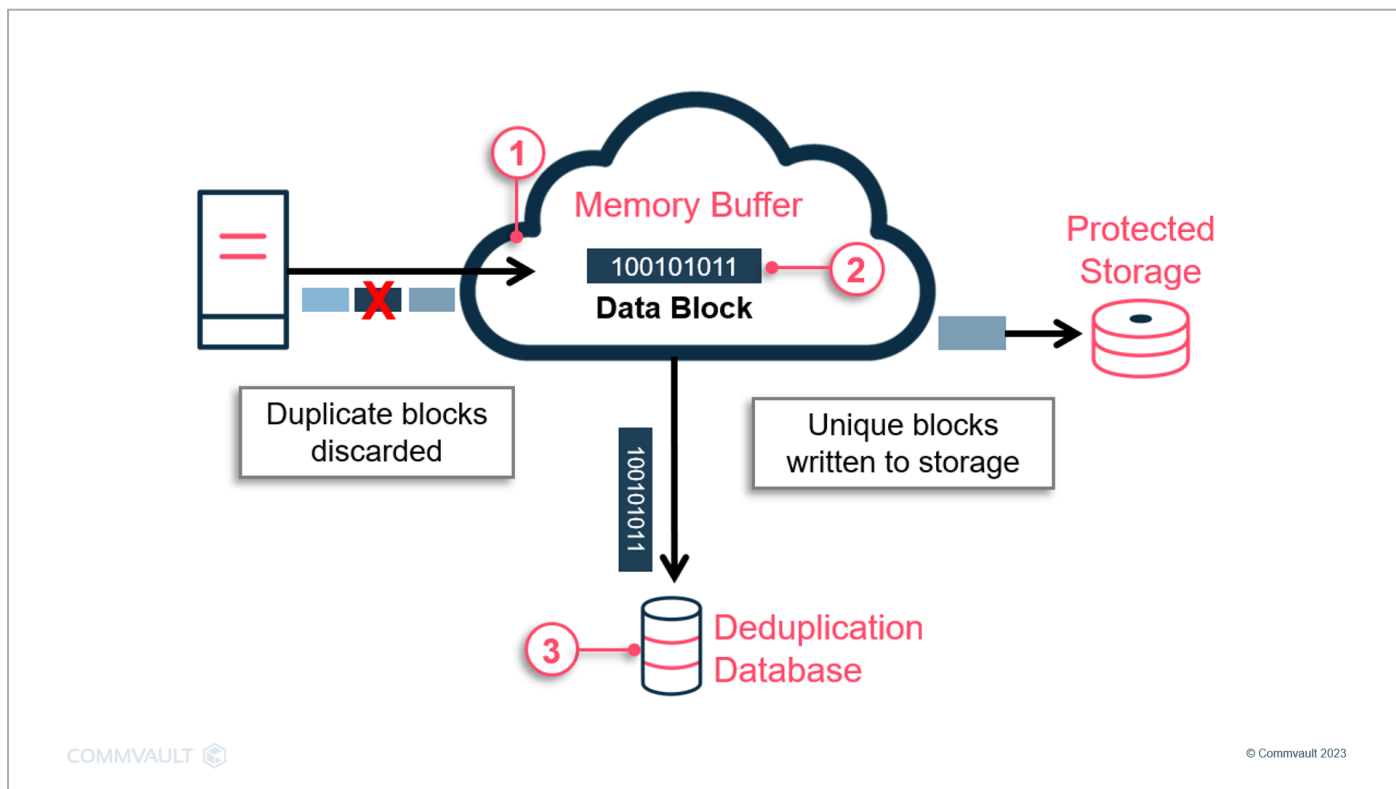
It is common for duplicate data to exist on storage-based media, networks, and virtual servers. Some examples include identical DLL files existing on different servers, or multiple users working on the same document, where each user modifies different blocks in the file while other blocks remain unchanged. Traditionally this redundant data is stored on disk or tape, which requires a significant amount of space. With Commvault deduplication storage techniques, a single copy of data and any subsequent references to the same data is stored only once, reducing the amount of space needed to save and protect it.

Benefits and Features

Commvault software has a unique set of deduplication features that are not available with most third-party deduplication solutions. Reduce storage and network resource requirements, shrink backup windows, efficiently copy data to off-site locations, and copy deduplicated data to tape, disk, or to a cloud environment.

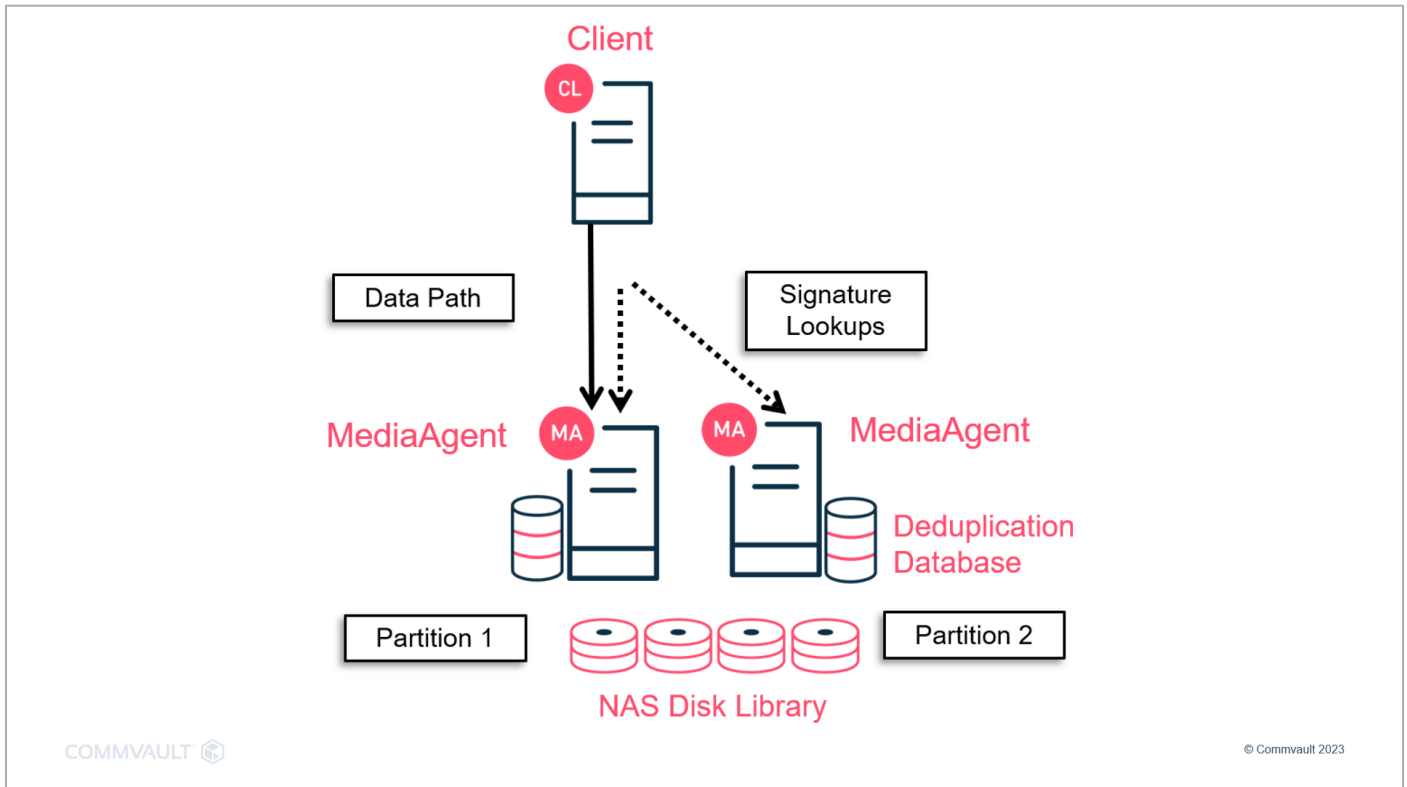
- **Efficient use of storage media** – Commvault deduplication provides processes that are used to efficiently move large amounts of data.
- **Efficient use of network bandwidth** – Client-side deduplication is used to deduplicate block data before it leaves the client. From that point forward, only changed blocks are sent over the network. This greatly reduces network bandwidth requirements after the first successful full backup is complete.
- **Fast Synthetic Full operations** – A Synthetic Full backup synthesizes full backup operations by copying previously backed up data into a new full backup job. Deduplication Accelerate Streaming Hash (DASH) Full backup reduces the time to perform Synthetic Full and traditional full backup operations. The DASH Full runs as a read-optimized Synthetic Full operation, which does not require traditional full backups to be performed. Once the first full backup has completed, blocks that have changed are protected during incremental or differential backups. A DASH Full runs in place of a traditional full or Synthetic Full, does not require movement of data, and updates the index files and deduplication database (DDB) when a full backup has completed.
- **Fast Auxiliary Copy Operations** – The DASH Copy operations are optimized auxiliary copy jobs that require only modified blocks to be sent to a second disk target. Because secondary copies do not require high bandwidth requirements, this is an ideal solution for sending off-site copies to secondary disaster recovery facilities.
- **Resilient Indexing and Restoration** – Although the deduplication database (DDB) checks signature hashes for deduplication purposes, it is not required during restore operations. Instead, the standard indexing methodology is used. This includes using the index directory and index files written at the conclusion of the job. This resiliency ensures that deduplicated data is restored even during unforeseen events, such as disaster recovery.

Deduplication Process and Data Protection



Production data is read from the source location and written into a memory buffer. This memory buffer is filled based on the defined block size. Note that the block size is referred to as a data block with a default of 128KB. A signature is then generated on the data block. The signature uniquely represents the bit makeup of the block. The signature is compared in the DDB to determine if the data block already exists.

Partitioned Deduplication Database Overview



Partitioned deduplication provides higher scalability and deduplication efficiency by allowing more than one Deduplication Database (DDB) partition to exist within a single deduplication engine. It works by logically dividing signatures between multiple databases. If two deduplication partitions are used, it effectively doubles the size of the deduplication store. Commvault software supports up to four database partitions.

During data protection jobs, partitioned DDBs and the data protection operation work using the following logic:

1. The signature is generated at the source. For primary data protection jobs using client-side deduplication, the source location is the client. For auxiliary DASH copy jobs, the source MediaAgent generates signatures.
2. Based on the generated signature it is sent to its respective database. The database compares the signature to determine if the block is duplicate or unique.
3. The defined storage plan data path is used to protect data. Regardless of which database the signature is compared in, the data path remains consistent throughout the job.

Partitioned Databases and Network-Attached Storage (NAS)

If partitioned deduplication is going to be implemented using two MediaAgents, use a shared disk library with a Network-Attached Storage (NAS) device. The NAS storage allows either MediaAgent to recover data even if the other MediaAgent is not available.

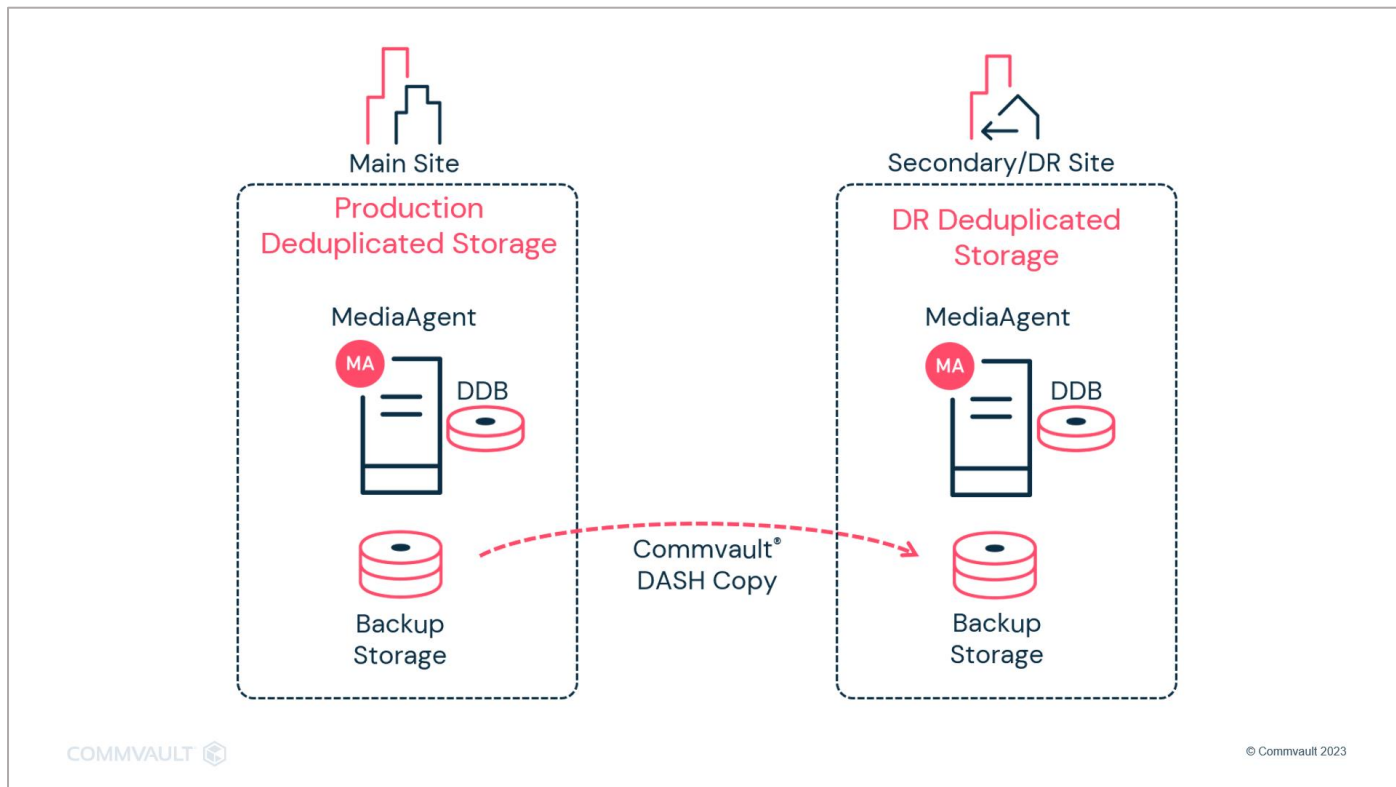
Partitioned Database for Scalability

The primary purpose for partitioned DDBs is to provide higher scalability. By balancing signatures between database partitions, you can scale up the size of a single deduplication store. If you have two partitions, the size of the store doubles; four partitions quadruples its size.

Partitioned Database for Resiliency

Using partitioned databases ensures resiliency. For instance, if one MediaAgent hosting a DDB goes offline, the other MediaAgent continues data protection jobs as the available DDB continues signature lookups. However, with the loss of one database, all signatures previously managed by the offline database would now be looked up in the remaining online database. This causes existing signatures managed in the offline database to be compared in the online database, which results in the signatures being treated as unique and additional data being written to the library.

Enable Deduplication



When configuring storage, you can enable deduplication with a simple click. The deduplicated storage can then be selected in a plan so that all subclients protected by this plan will have their datasets stored in the same storage and will deduplicated against each other. When enabling deduplication, the location for the DDB partition must be defined. If more than one MediaAgent is sharing the storage unit (NAS or cloud), up to four partitions can be created for scalability.

The DASH Copy operations are optimized auxiliary copy jobs that require only modified blocks to be sent to a second disk target. Deduplication can be enabled for primary storage as well as secondary storage. When enabling deduplication on a secondary copy, the data transfer leverages deduplication by sending only the new blocks. All existing blocks are simply dropped from the source, avoiding unnecessary transmission of data.

Configure Deduplication for Storage

The screenshot displays the Commvault web interface. On the left, the navigation menu shows 'Storage' expanded, with 'Disk' selected (indicated by a red circle 1). The main content area is titled 'Disk' and contains a table of storage configurations. In the top right of this area, an 'Add' button is highlighted with a red circle 2. A modal window titled 'Add Deduplication DB location' is open in the foreground. Within this modal, the 'MediaAgent' dropdown is set to 'mediaagent1' (marked with a red circle 3), and the 'Deduplication DB location' text field contains 'F:\Disk Target DDB'. The 'ADD' button at the bottom right of the modal is marked with a red circle 4.

Name ↑	Status	Capacity	Free space	Actions
Basic Disk	Online	25 GB	17.22 GB	⋮
Disk Target	Online	25 GB	17.22 GB	⋮
Production Storage	Online	50 GB	36.92 GB	⋮

Deduplication is only available during the storage initial configuration and cannot be turned on later. It is important to collect information on the location of the DDB. It should be a dedicated set of disks, preferably SSD, located on the same MediaAgent to which the storage is configured.

1. Select **Storage > Disk** to view configured disk storage
2. Click **Add** to create new disk storage
3. Enter a name
4. Click **ADD** to configure the storage target

Configure Deduplication for Storage ...cont'd

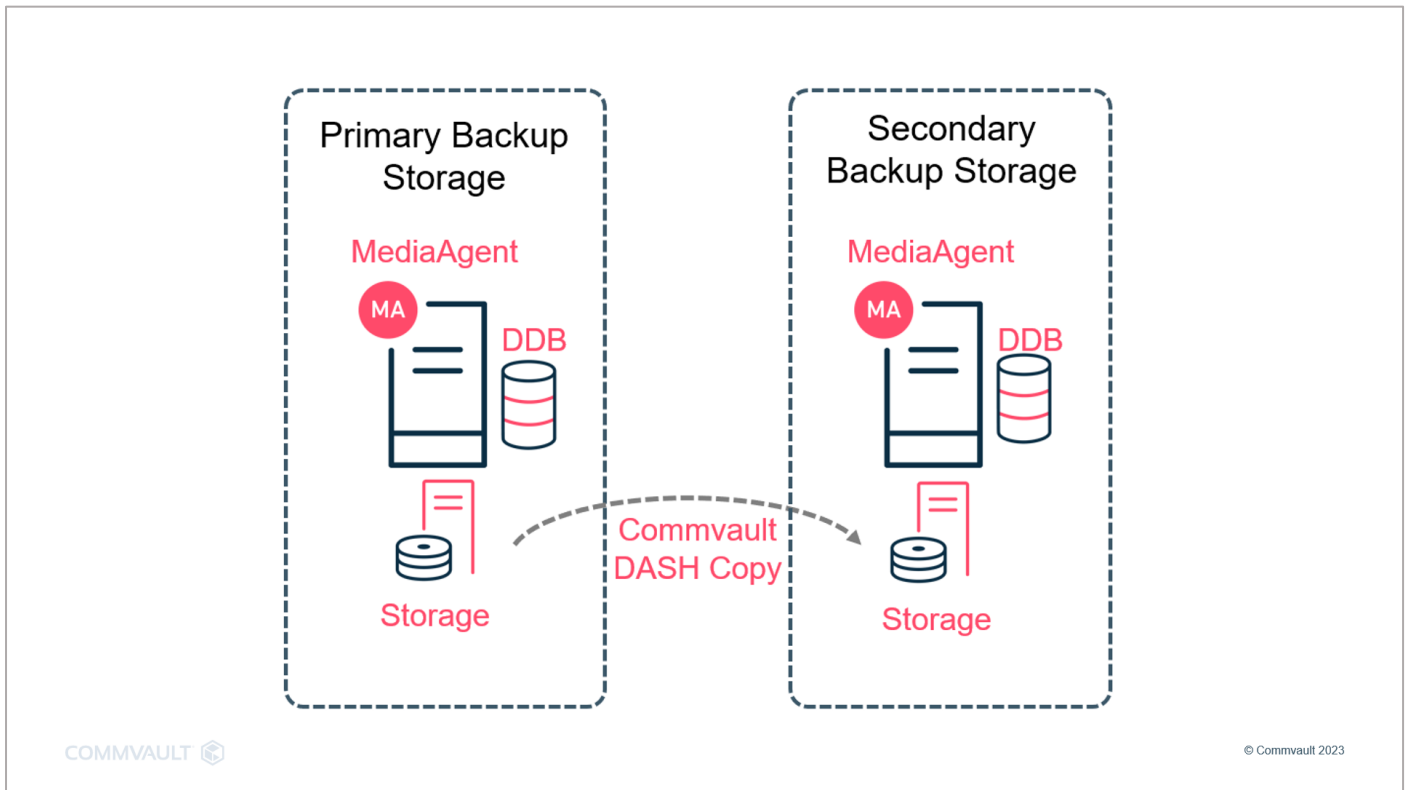
The screenshot shows the 'Add disk storage' configuration window in the Commvault interface. The window has a sidebar on the left with navigation options like 'Guided setup', 'Dashboard', 'Protect', 'Data Insights', 'Auto recovery', 'Jobs', 'Reports', 'Monitoring', 'Storage', 'Manage', 'Developer tools', 'Workflows', and 'Web console'. The main area is titled 'Add disk storage' and contains the following fields and controls:

- Name ***: A text field with the value 'Disk Target'.
- Disk Target**: A text field.
- Backup location**: A section with an 'Add' button. It contains a table with two columns: 'MediaAgent' and 'Backup location'. The first row shows 'mediaagent1' and 'G:\Disk Target'. Red circle 5 is on the 'mediaagent1' dropdown, and red circle 6 is on the 'G:\Disk Target' text field.
- Use deduplication**: A toggle switch that is currently turned on. Red circle 7 is on the toggle.
- Deduplication DB location**: A section with an 'Add' button. It contains a table with two columns: 'MediaAgent' and 'Deduplication DB location'. The first row shows 'mediaagent1' and 'F:\Disk Target DDB'. Red circle 8 is on the 'F:\Disk Target DDB' text field.
- EQUIVALENT API**: A text field.
- CANCEL** and **SAVE** buttons: At the bottom right. Red circle 9 is on the 'SAVE' button.

5. Select the MediaAgent with access to the storage
6. Browse or enter the path of the first storage mount path
7. Toggle on **Use deduplication** to enable deduplication
8. Browse or type the path to the location to host the deduplication database
9. Click **SAVE**

Plans

Plans Overview



The configuration of data protection in the Commvault Command Center is achieved by creating a server protection plan consisting of several configuration options easily set in a single wizard. Options include storage, retention, backup schedule (RPO), and offsite copy configuration. Different plans can be created, each used for a specific goal.

- **Server plan** – Define data protection settings for file system and application agents
- **Exchange plan** - Configure the Exchange Mailbox agent
- **Data classification plan** – Configure the indexing of the content of backup data (content indexing)

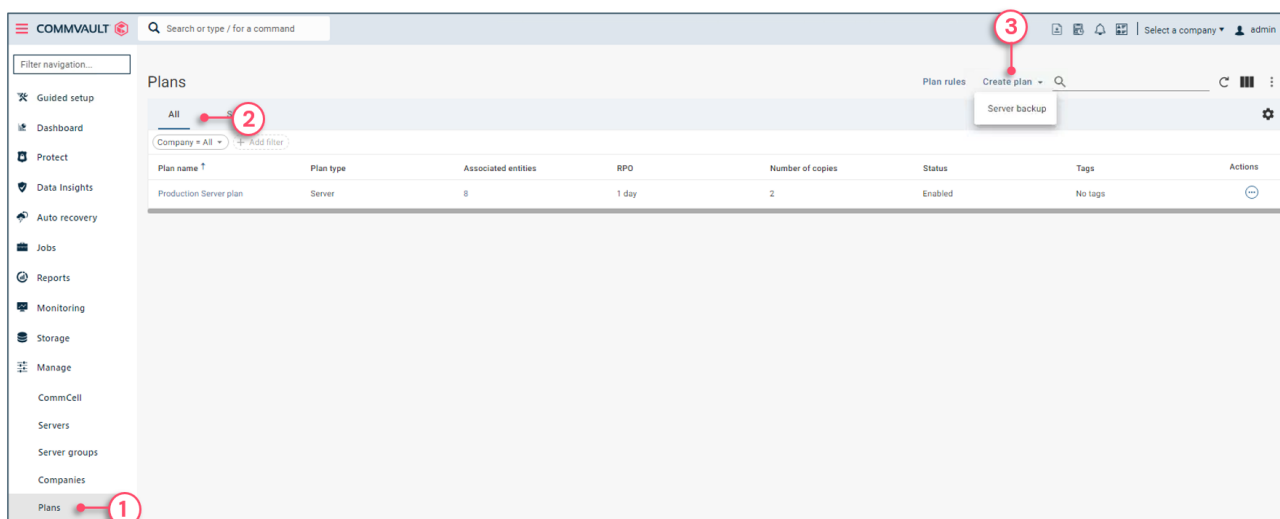
Any plan can be used as a base plan and a derived plan can be created from the base plan. The derived plan inherits the same protection options, which can then be overridden. When modifying the base plan, changes are also applied to derived plans unless they are overridden.

Scenario: Using plan templates

You have multiple servers for which protection requirements are similar except for one option, such as the backup frequency.

Solution: Create a base plan with all settings defined and set the RPO option to 'Override must'. Use the plan as a template from which administrators will create derived plans and set the backup frequency.

Configure a Server Plan



A server plan protects servers using the file system agent. Supported operating systems include Microsoft Windows, UNIX/Linux, and Mac OS. A server plan can be used to protect virtual machines protected with the Virtual Server Agent (VSA), and to configure application agents except for the Exchange Mailbox agent, which is configured using an Exchange plan. Finally, if the file system, virtual machine, or application requires protection using hardware snapshot, Commvault IntelliSnap can be configured under the *Snapshot* section of the server plan.

1. Select **Manage > Plans**
2. This window displays the existing plans
3. Click **Create plan** and then click **Server backup**

Configure a Server Plan ...cont'd

The screenshot shows the 'Create Server Backup Plan' wizard in the Commvault console. The 'Backup Destinations' step is active. A modal window titled 'Add copy' is open, allowing configuration of a new backup copy. The modal includes fields for 'Name' (set to 'Primary'), 'Storage' (set to 'Production Storage'), 'Retention rules' (set to 1 month), and a table for 'Rules' (set to Weekly Fulls every 90 days). Red circles with numbers 4, 5, and 6 are overlaid on the 'ADD COPY' button, the 'Name' field, and the 'Storage' dropdown respectively.

4. Click **ADD COPY** to select storage for the primary copy of backup data

5. Configure the primary copy details

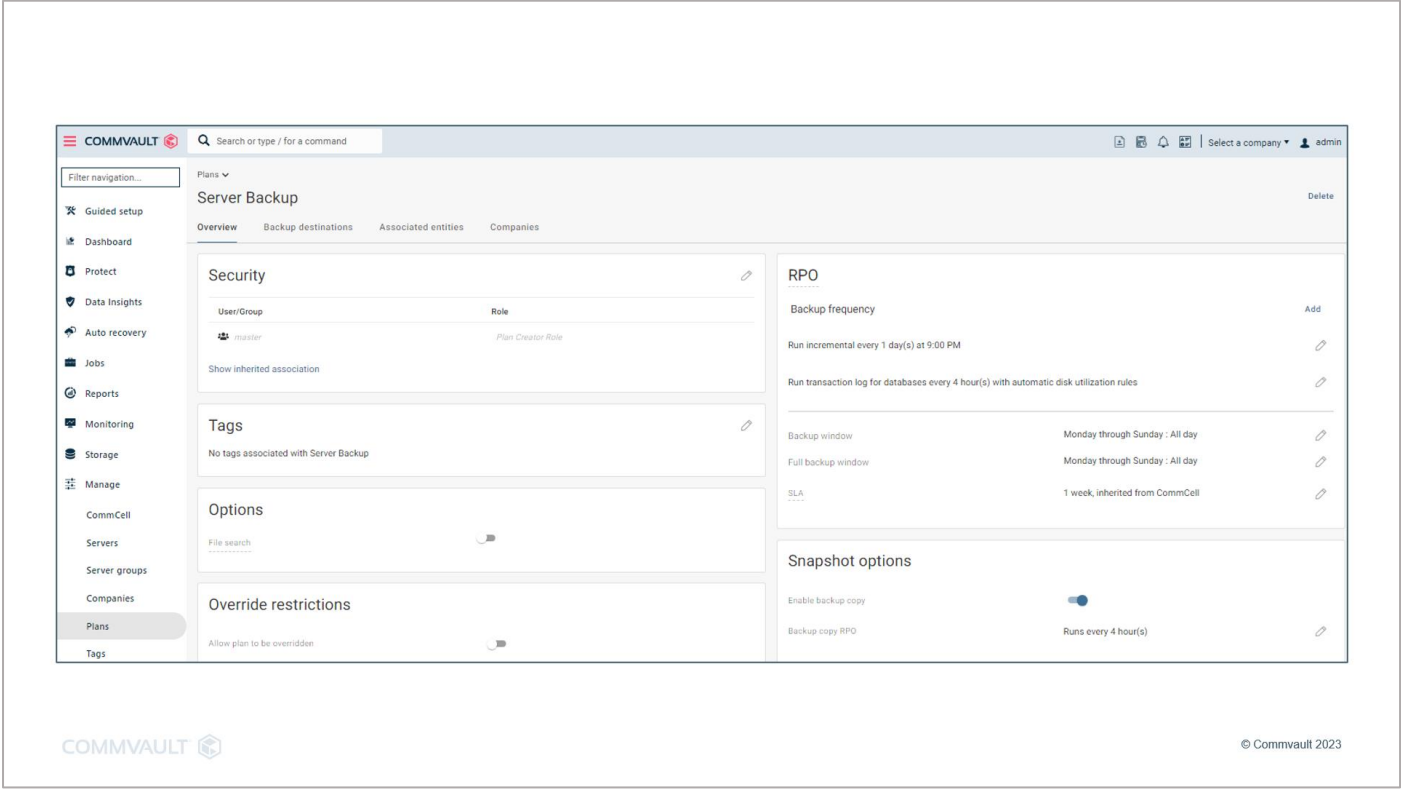
6. Select the storage to be used, configure the required retention and apply any necessary extended retention rules

Configure a Server Plan ...cont'd

The image displays two screenshots of the Commvault 'Create Server Backup Plan' wizard. The top screenshot shows the 'Recovery Point Objective' step, where users configure backup frequency, transaction log frequency, backup window, and full backup window. Red callout 7 points to the frequency field, and red callout 8 points to the backup window field. The bottom screenshot shows the 'Options' step, where users configure snapshot options, retention period, number of snapshot recovery points, enable backup copy, and backup copy RPO. Red callout 9 points to the 'SUBMIT' button. Both screenshots show a sidebar with navigation options like Guided setup, Dashboard, Protect, Data Insights, Auto recovery, Jobs, Reports, Monitoring, Storage, and Manage.

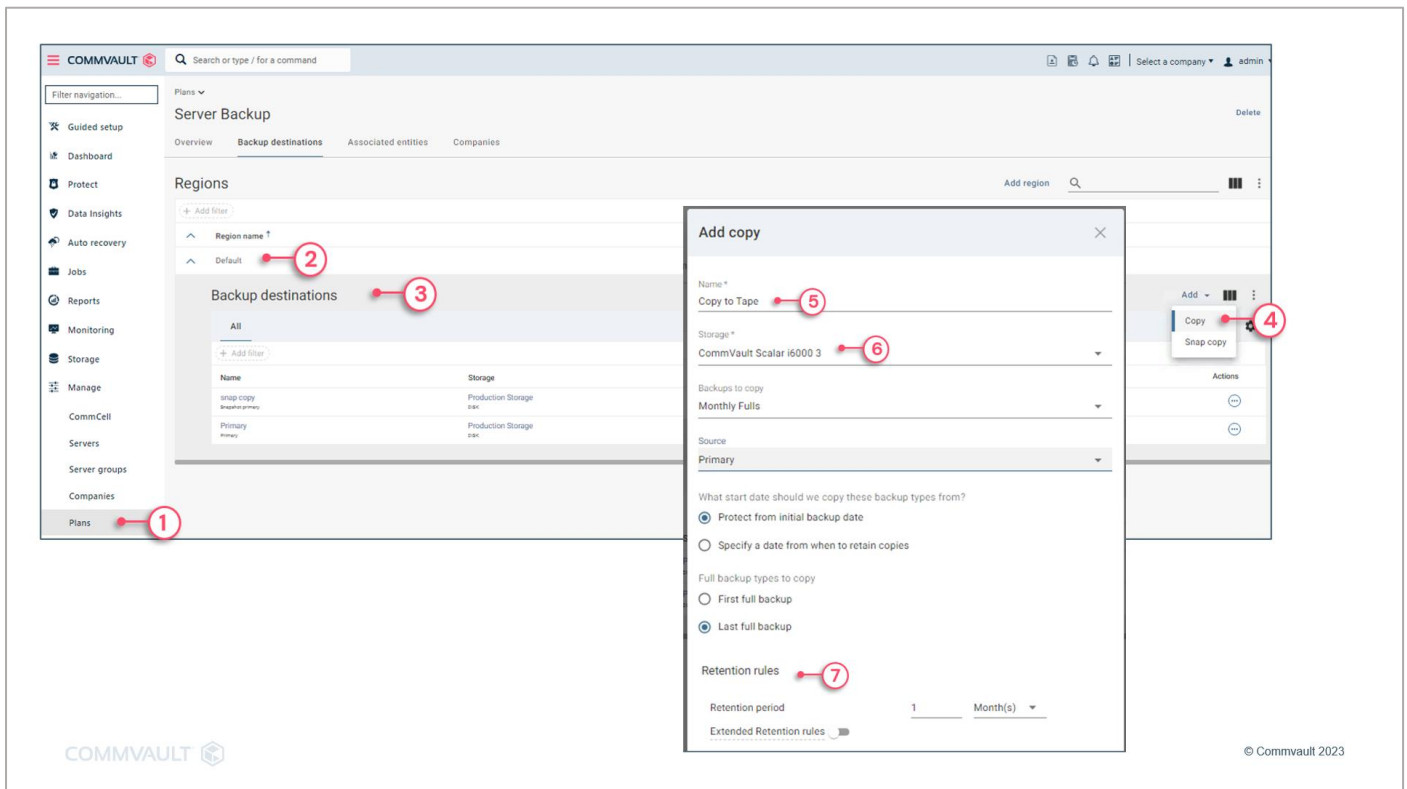
7. Configure the Recovery Point Objective, or backup frequency
8. Configure backup windows
9. Configure any advanced options for snapshot copies and click **SUBMIT**

Configure a Server Plan ...cont'd



Once complete, you are shown the Plan overview page.

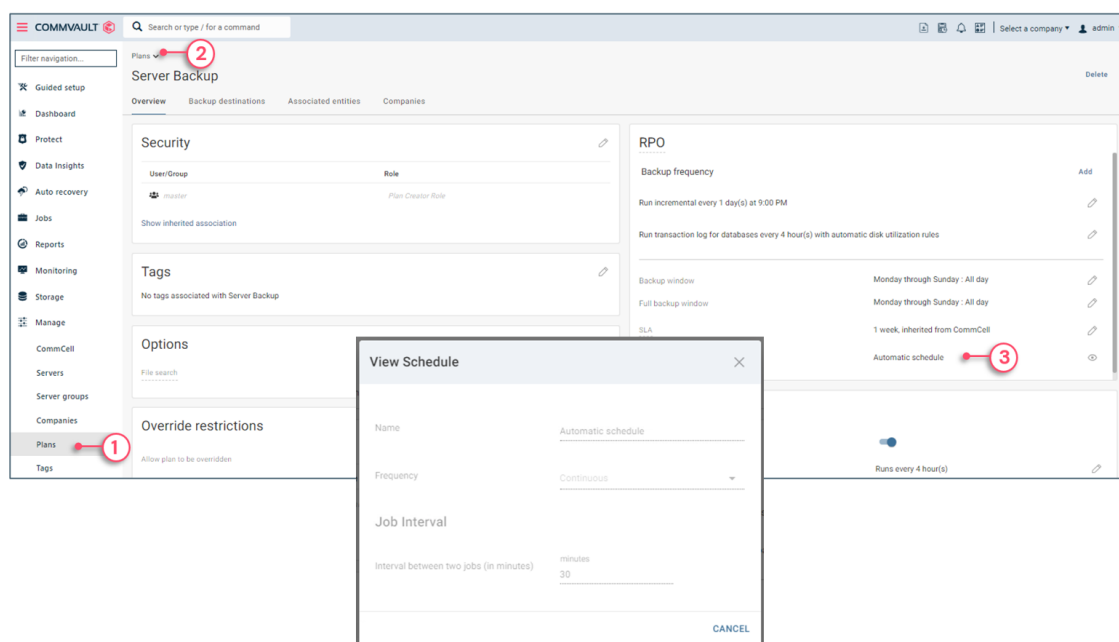
Secondary and Auxiliary Copies



A secondary copy is used to synchronize protected data with a source copy. All full, incremental, differential, transaction log, or archive jobs from a source copy are also managed by the secondary copy. Secondary copies are useful when you want a consistent point-in-time copy of all protected data available for restore. It is recommended to have at least one offsite secondary copy to ensure full recovery of data, should a disaster occur. Secondary copies can be disk, cloud, or tape.

1. Select **Manage > Plans**
2. Click the plan name
3. The *Backup destinations* section displays copies currently configured
4. Click **Add** and then click **Copy** to add a copy to the plan
5. Enter a name for the secondary copy
6. Choose the storage
7. Set the retention period and click **SAVE** to create the secondary copy

Secondary Copy Schedule



A built-in secondary copy schedule is created when adding a secondary copy to a plan. By default, the system validates if new jobs exist in the primary copy that were not copied to the secondary copy in 30-minute increments. New jobs are copied once they are created. If there are no new jobs, the auxiliary copy job completes without copying data. It is recommended to keep the default schedule to ensure the best synchronization possible between copies; it can be modified if needed.

1. Select *Manage > Plans*
2. Click the plan name
3. Click the secondary copy schedule name

Modify the schedule frequency to meet your needs.